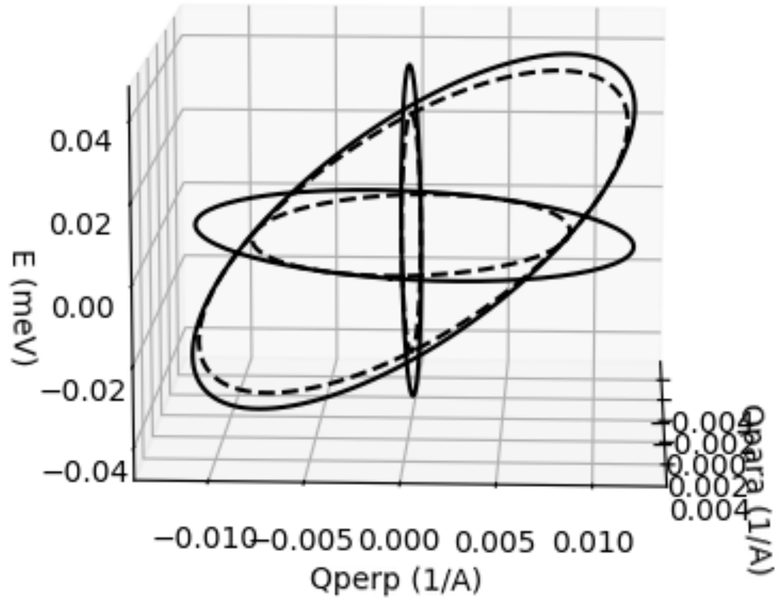
Resolution Function of a Triple-Axis Spectrometer

Bing Li
Computational Instrument Scientist in the Triple-Axis Group
Oak Ridge National Laboratory

02/04/2025

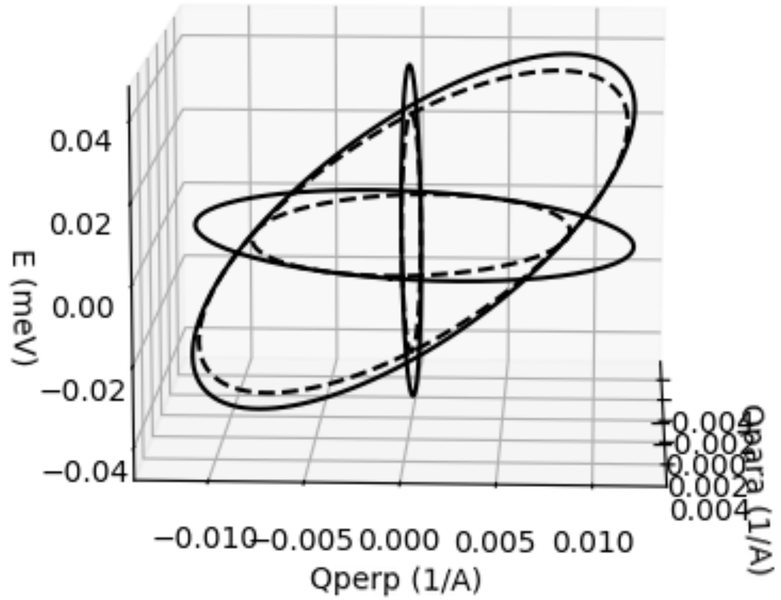
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- A measurement by the Triple-Axis Spectrometer
  - Intensity
  - Broadening from instrument
- Resolution function from Cooper-Nathans's method
  - How to calculate the resolution matrix M
  - M described by an ellipsoid
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  - Diffraction: integrated intensity and the Lorentz factor
  - Diffraction: deconvolution for correlation length
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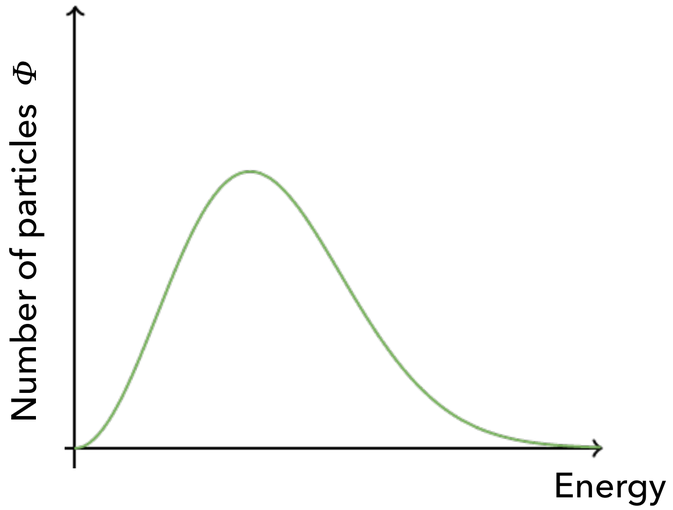
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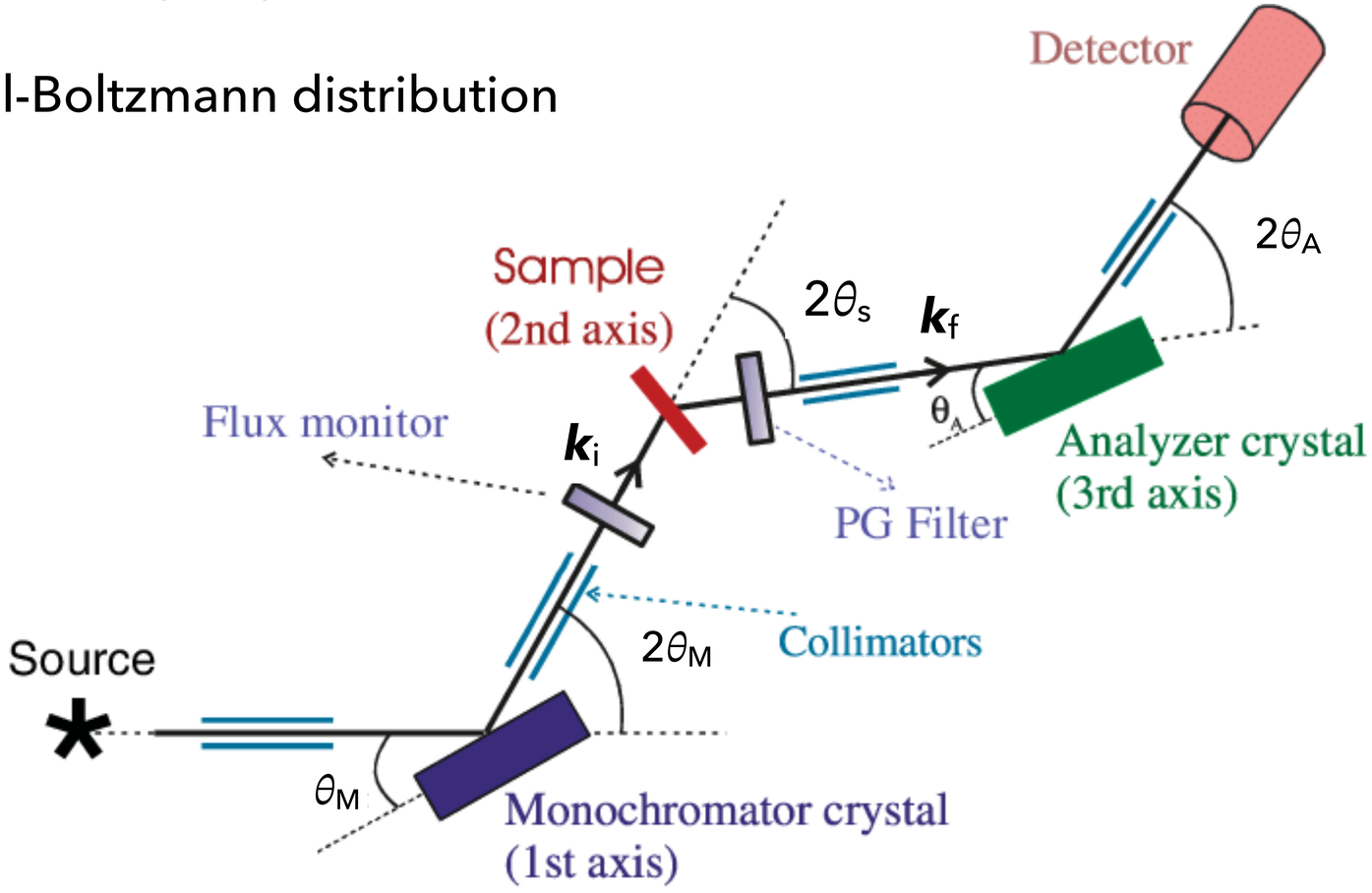
- A measurement by the Triple-Axis Spectrometer, set up to measure at point ($\boldsymbol{Q}$, E)
  - Intensity -> Normalization factor R_0 of the resolution function
  - Broadening from instrument -> FWHM of the resolution function
- Resolution function from Cooper-Nathans's method
  - How to calculate the resolution matrix M
  - M described by an ellipsoid
- Applications
  - Diffraction: integrated intensity and the Lorentz factor
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When We Set Up TAS To Measure at Point (Q,E)...

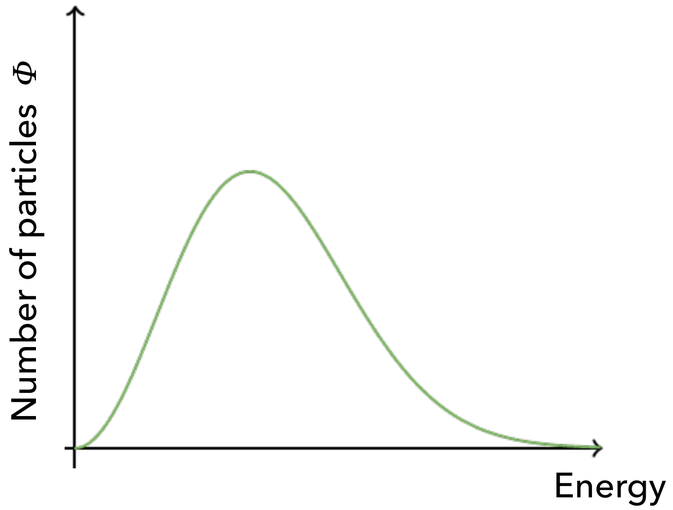
- **Source**: Moderated neutron follows Maxwell-Boltzmann distribution

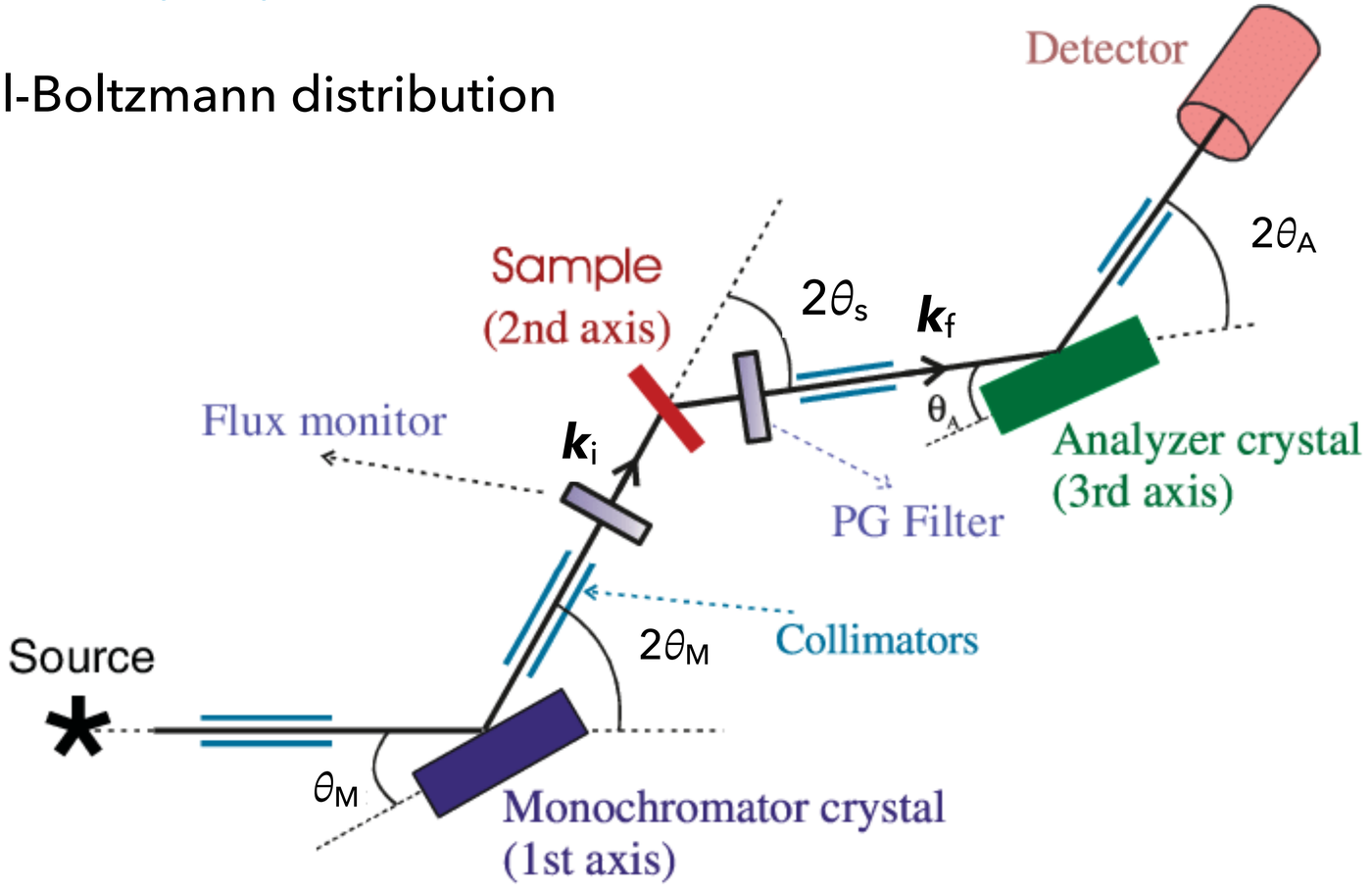




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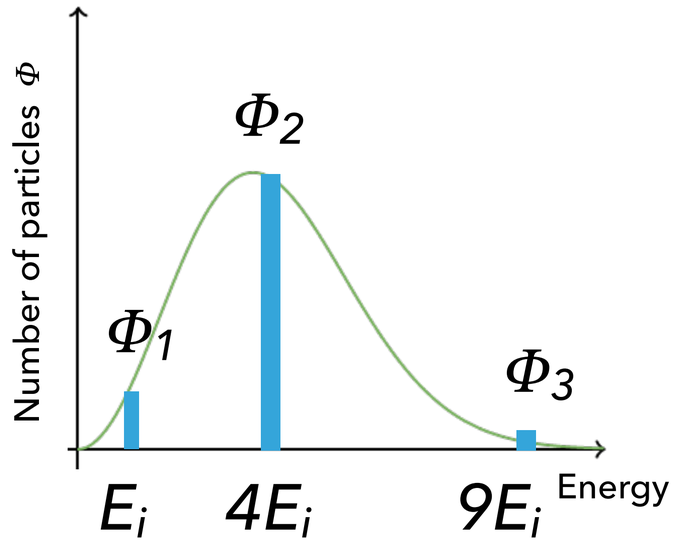


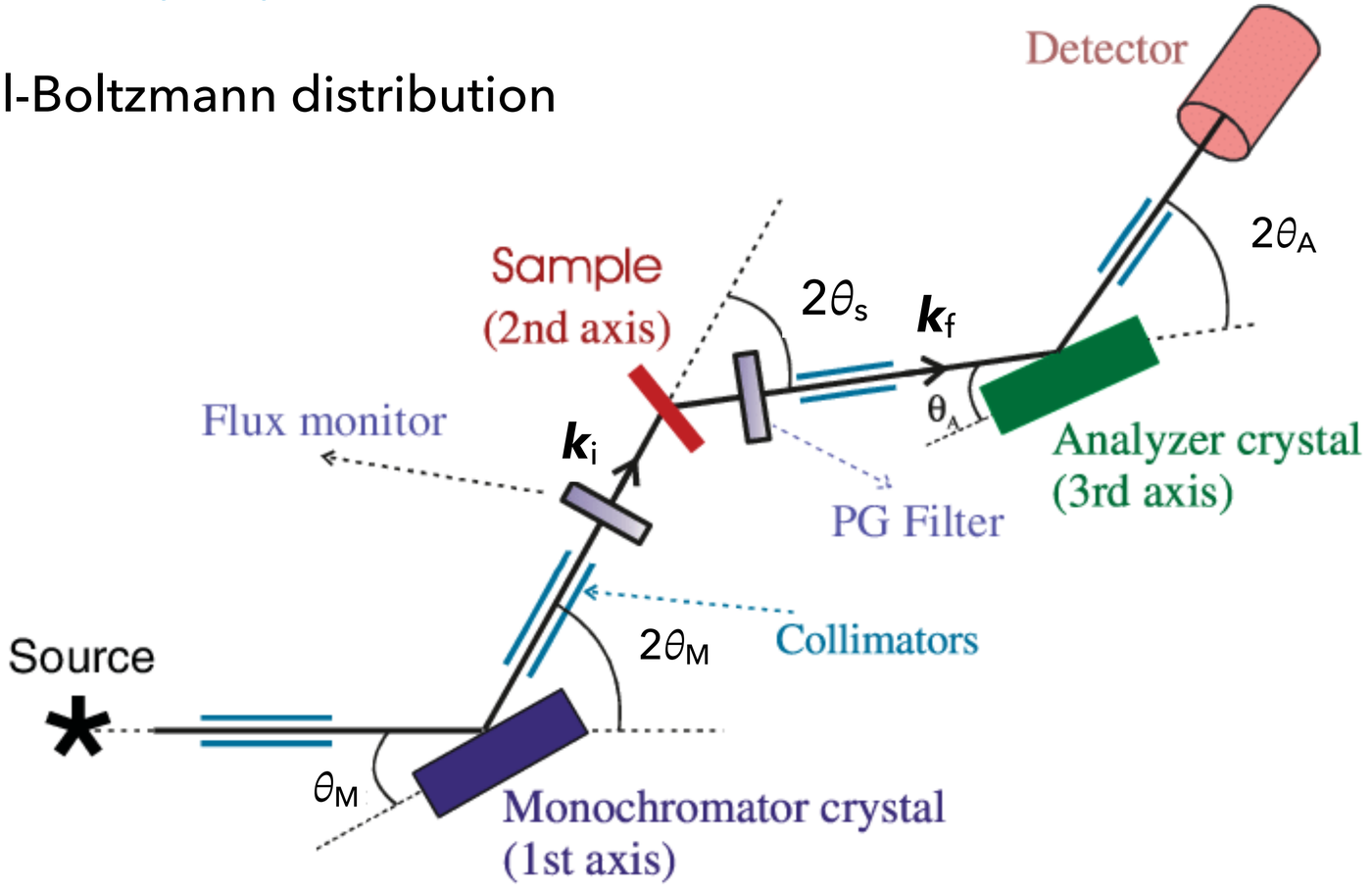


- **Mono**: select wavelength according to the Bragg's law, $2d\sin(\theta_s) = n\lambda$
- Beam contains $\boldsymbol{k}_i$ and its harmonics ($2\boldsymbol{k}_i$, $3\boldsymbol{k}_i$ ...), $\Phi = \Phi_1 + \Phi_2 + \Phi_3$

When We Set Up TAS To Measure at Point (Q,E)...

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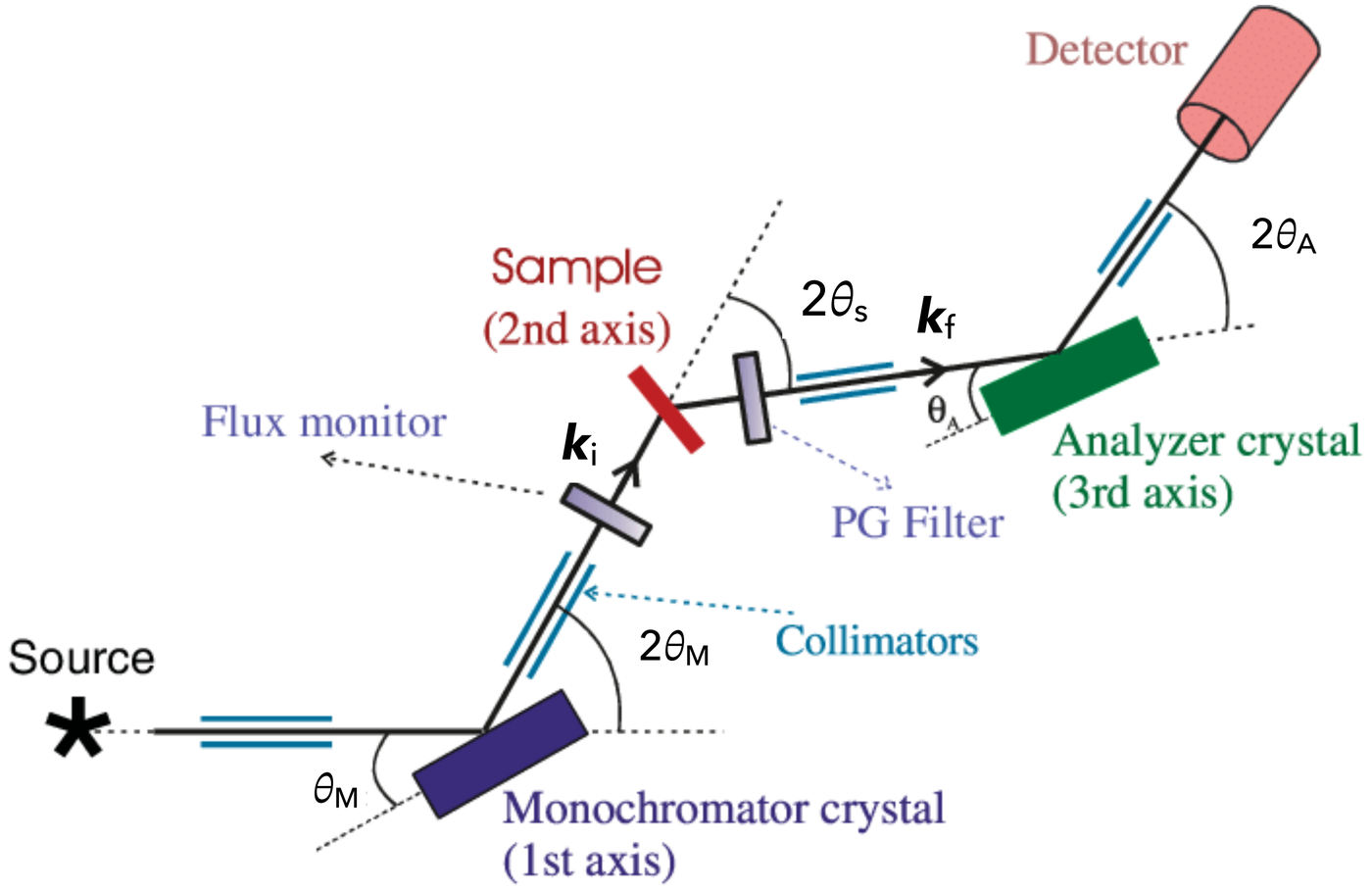




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Understanding the Intensity

- **Monitor**: efficiency $\eta \sim \frac{1}{v}$,
  $\Phi_{\text{monitor}} \sim \frac{1}{k_\text{i}} \left(\Phi_1 + \Phi_2/2 + \Phi_3/3\right)$
- **Sample**: dynamical structure factor $S(\mathbf{Q}, E)$
- **PG filter:** Transmission T_PG~0.8 (E_f~14 meV)
- **Analyzer**: reflectivity $R_A = R_A(k_f)\frac{k_f^3}{\tan(\theta_A)}$
- **Detector**: correction for non-linear response, $\Phi_\text{det} \sim \Phi_1 \cdot S(\mathbf{Q}, E) \cdot T_\text{PG} \cdot R_\text{A} \cdot \eta_\text{det}$



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- **Counts** measured in a given period of time $N \sim \Phi \cdot v$
- **Intensity**: Normalized detector counts

$$I(\mathbf{Q}, E) = \frac{k_f\ \Phi_{\text{det}}}{k_i\ \Phi_{\text{monitor}}}$$
$$\sim \frac{k_f \cdot \Phi_1 \cdot S(\mathbf{Q}, E) \cdot T_{\text{PG}} \cdot R_{\text{A}} \cdot \eta_{\text{det}}}{k_i \cdot (\frac{1}{k_i}\Phi_1 + \dots)}$$
$$\sim k_f \cdot S(\mathbf{Q}, E) \cdot (OtherFactors)$$

- $S(\mathbf{Q}, E)$ can be calculated from FullProf or Sunny

Understanding the Intensity

- **Monitor**: efficiency $\eta \sim \frac{1}{v}$,
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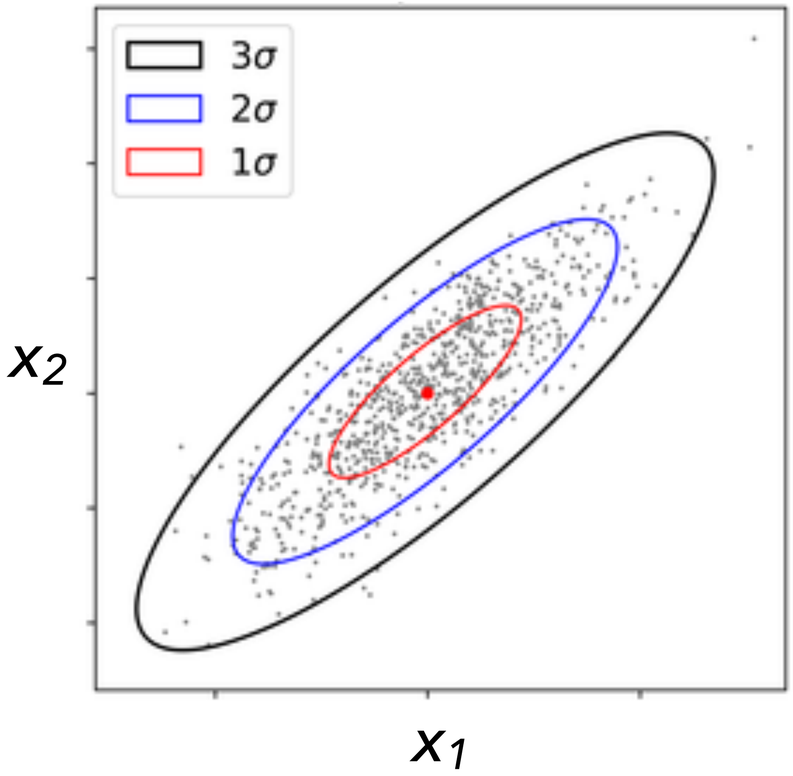
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$$\sim k_f \cdot S(\mathbf{Q}, E) \cdot (OtherFactors)$$

- $S(\mathbf{Q}, E)$ can be calculated from FullProf or Sunny
- Directly comparison btw experiment and simulation
  IF data is collected in the fixed-E_f mode
- **Monitor efficiency needs to be 1/k !!!**
- **Need to know the fraction of harmonic** $\Phi_1, \Phi 3, \dots$

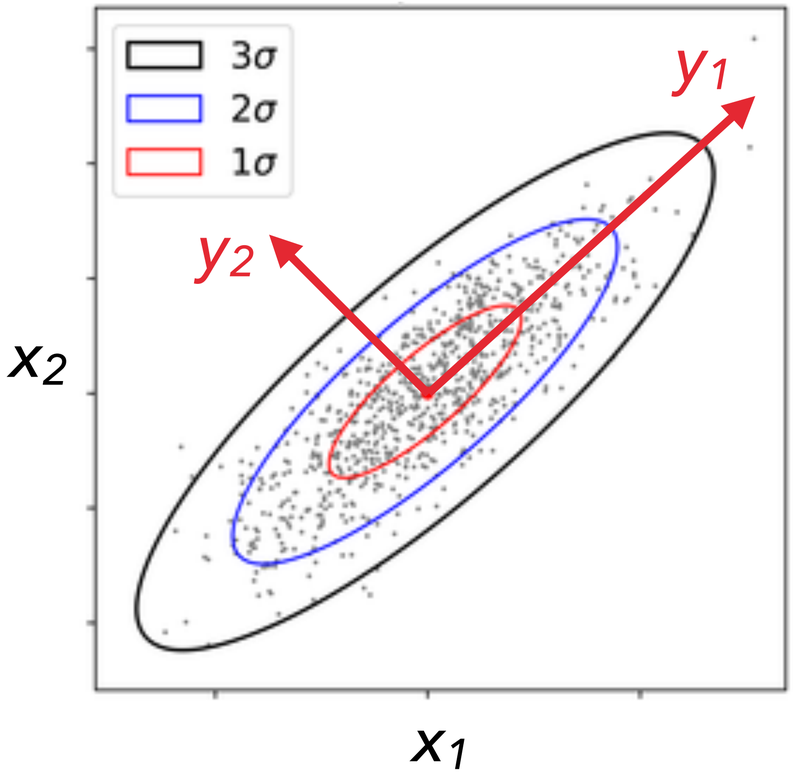
Broadening From the Instrument

- Real-life signals are not sharp like δ-functions, but follows a Gaussian distribution (Central limit theorem)
- Probability distribution $P(\mathbf{X} = (x_1, x_2, \dots)) \sim \exp\left(-\frac{1}{2}\mathbf{X}^T M \mathbf{X}\right)$



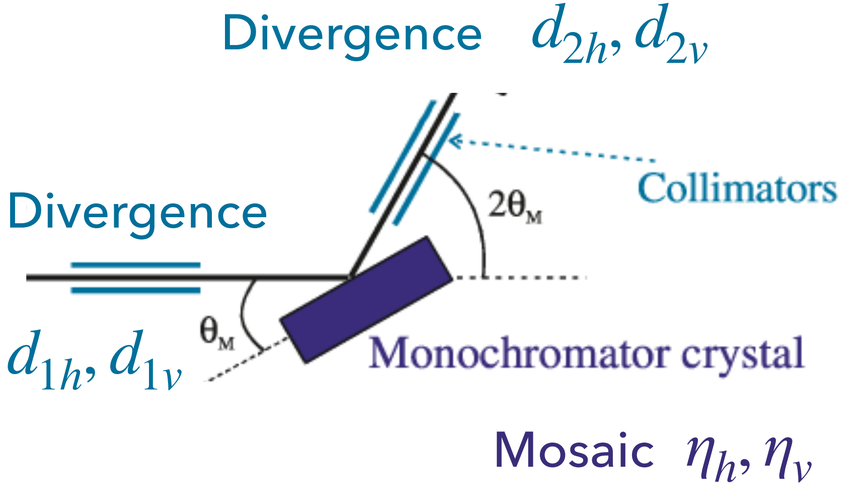
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- Probability distribution $P(\mathbf{X} = (x_1, x_2, \dots)) \sim \exp\left(-\frac{1}{2}\mathbf{X}^T M \mathbf{X}\right)$
- Find independent degree-of-freedoms $\mathbf{Y} = (y_1, y_2, \dots)$, $\mathbf{X} = A\mathbf{Y}$
- Conversion of matrix, $M = A^T D A$, with $D = \begin{bmatrix} d_1 & & \\ & d_2 & \\ & & \ddots \end{bmatrix}$
- $P(\mathbf{Y}) \sim \exp\left(-\frac{1}{2}\mathbf{Y}^T \begin{bmatrix} d_1 & & \\ & d_2 & \\ & & \ddots \end{bmatrix} \mathbf{Y}\right) = \exp(-\frac{1}{2} d_1 y_1^2) \cdot \exp(-\frac{1}{2} d_2 y_2^2) \dots$

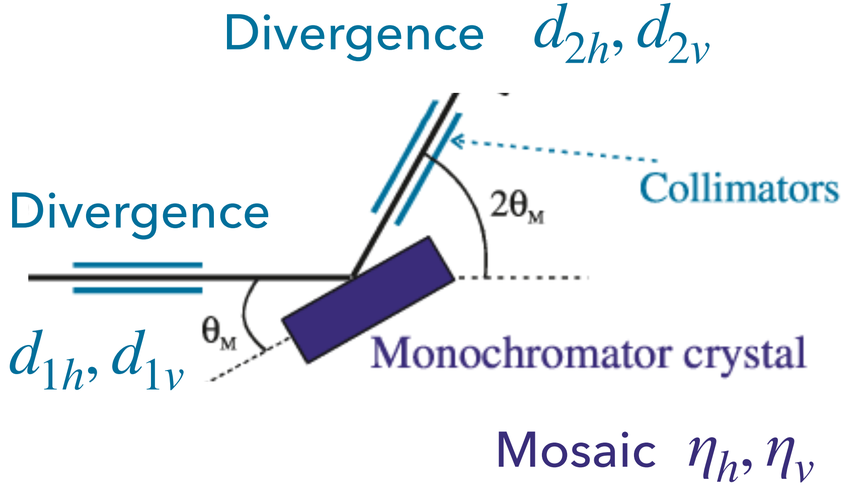


Broadening From the Instrument

- Example 1. After a Soller slit collimator, the beam can be described as $P(\delta_h, \delta_v) = \exp\left(-\frac{1}{2}\begin{bmatrix}\delta_h \\ \delta_v\end{bmatrix}^T \begin{bmatrix}1/d_h^2 & \\ & 1/d_v^2\end{bmatrix}\begin{bmatrix}\delta_h \\ \delta_v\end{bmatrix}\right)$, where ($\delta_h$, δ_v) are horizontal/vertical angular beam deviation, (d_h, d_v) are the divergence of the collimator



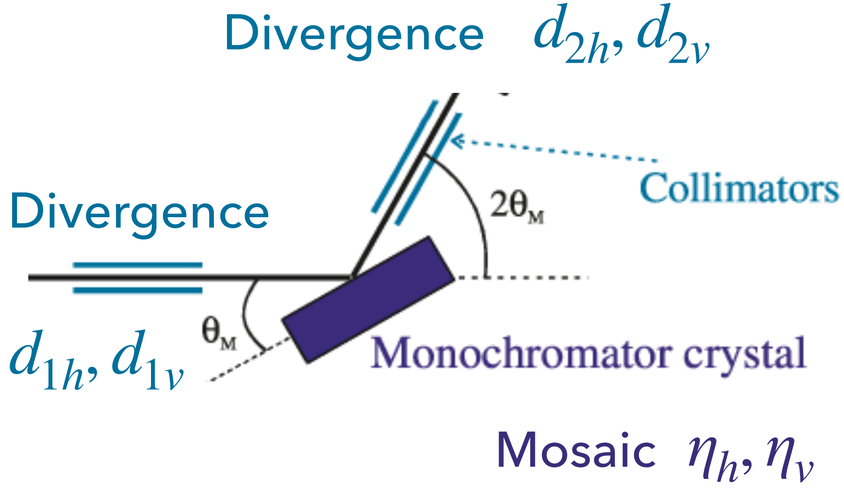
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- Example 2. After a monochromator, the beam can be described as $P(\xi_h, \xi_v) = \exp\left(-\frac{1}{2}\begin{bmatrix}\xi_h\\ \xi_v\end{bmatrix}^T \begin{bmatrix}1/\eta_h^2 & \\ & 1/\eta_v^2\end{bmatrix}\begin{bmatrix}\xi_h\\ \xi_v\end{bmatrix}\right)$, where ($\xi_h$, ξ_v) are horizontal/vertical angular beam deviation, (η_h, η_v) are the mosaic spread of the monochromator

Broadening From the Instrument



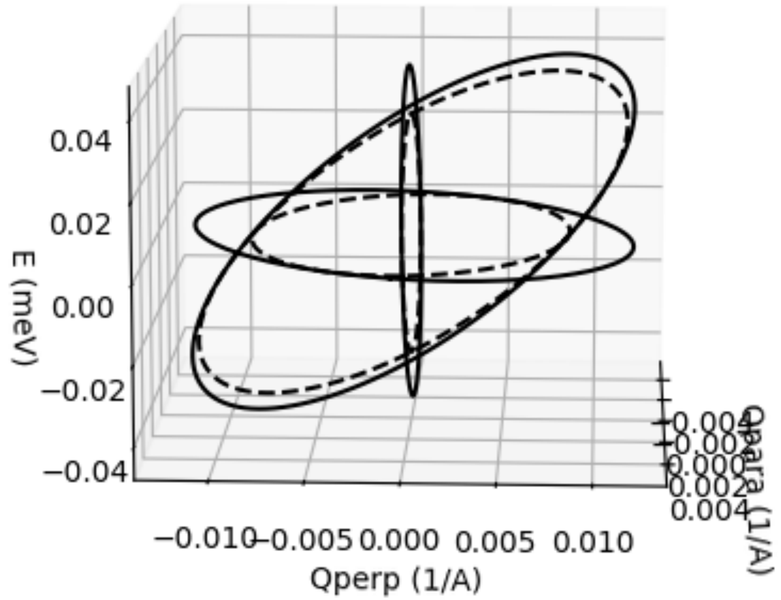
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- Note that the independent DoFs (δ_h, δ_v) and (ξ_h, ξ_v) are not variable we care, need to change the variables to (**Q**, E)

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Resolution Function From Cooper-Nathans's Method

- Resolution function $R(\mathbf{Q},E) = R_0 \cdot \frac{\sqrt{|M|}}{4\pi^2} \exp\left(-\frac{1}{2}\begin{bmatrix}\mathbf{Q}\\E\end{bmatrix}^T M \begin{bmatrix}\mathbf{Q}\\E\end{bmatrix}\right)$
- Resolution matrix *M*
- Covariance matrix $M^{-1} = BA(G + C^T F C)^{-1} A^T B^T$
- Normalized to R_0 instead of 1, $\iint d\mathbf{Q}\, dE\; R(\mathbf{Q},E) = R_0$
- $R_0 = \frac{\pi^2}{4} \frac{k_i^3 \cot\theta_M R_M}{\sin\theta_M} \frac{k_f^3 \cot\theta_A R_A}{\sin\theta_A} \frac{\sqrt{|F|}}{\sqrt{|G + C^T F C|}}$, where R_M and R_A are integrated reflectivity of mono and analyzer, respectively

$$g(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp(-\frac{x^2}{2\sigma^2})$$

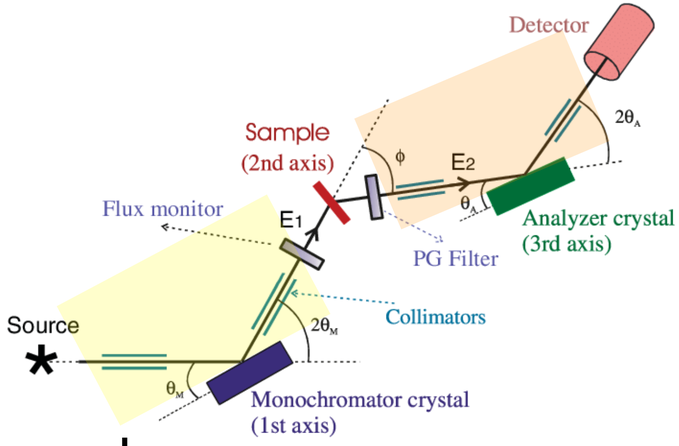
$$M = 1/\sigma^2$$

$$M_{ij}^{-1} = \left\langle \sigma_i \sigma_j \right\rangle$$

PoPovici, Acta Cryst. (1975). A31, 507

Matrix Elements In M Inverse $M^{-1} = BA(G + C^T F C)^{-1} A^T B^T$

- Horizontal and vertical angular divergence γ_i and δ_i after collimator C_i (i = 0,1, 2, 3), $\mathbf{U} = (\gamma_0, \gamma_1, \delta_0, \delta_1, \gamma_2, \gamma_3, \delta_2, \delta_3)$
- Horizontal and vertical angular divergence ξ and ξ' after mono and analyzer $\mathbf{W} = (\xi_M, \xi'_M, \xi_A, \xi'_A)$
- $P(\mathbf{U}, \mathbf{W}) = P_0 \frac{\sqrt{|G|}}{16\pi^4} \exp(-\frac{1}{2}\mathbf{U}^T G \mathbf{U}) \frac{\sqrt{|F|}}{4\pi^2} \exp(-\frac{1}{2}\mathbf{W}^T F \mathbf{W})$,
- where $G = \begin{bmatrix} \alpha_0^{-2} & & & & & & & \\ & \alpha_1^{-2} & & & & & & \\ & & \beta_0^{-2} & & & & & \\ & & & \beta_1^{-2} & & & & \\ & & & & \alpha_2^{-2} & & & \\ & & & & & \alpha_3^{-2} & & \\ & & & & & & \beta_2^{-2} & \\ & & & & & & & \beta_3^{-2} \end{bmatrix}$ and $F = \begin{bmatrix} \eta_M^{-2} & & & \\ & \eta_M'^{-2} & & \\ & & \eta_A^{-2} & \\ & & & \eta_A'^{-2} \end{bmatrix}$



PoPovici, Acta Cryst. (1975). A31, 507

Matrix Elements In M Inverse $M^{-1} = BA(G + C^T F C)^{-1} A^T B^T$

- Notice the angular divergences
  $\mathbf{U} = (\gamma_0, \gamma_1, \delta_0, \delta_1, \gamma_2, \gamma_3, \delta_2, \delta_3)$ and $\mathbf{W} = (\xi_M, \xi'_M, \xi_A, \xi'_A)$ are not independent, but tied by the Laue condition, $\mathbf{Q} = \mathbf{k_i} - \mathbf{k_f}$, giving the constraint $\mathbf{W} = C\mathbf{U}$

$$C = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & & & & & & \\ & & \frac{1}{2\sin\theta_M} & \frac{-1}{2\sin\theta_M} & & & & \\ & & & & \frac{1}{2} & \frac{1}{2} & & \\ & & & & & & \frac{1}{2\sin\theta_A} & \frac{-1}{2\sin\theta_A} \end{bmatrix}$$

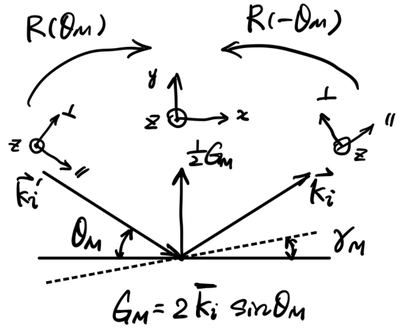
- $P(\mathbf{U}) = \dfrac{P_0}{16\sin\theta_M \sin\theta_A} \dfrac{\sqrt{|G|}}{16\pi^4} \dfrac{\sqrt{|F|}}{4\pi^2} \exp(-\frac{1}{2}\mathbf{U}^T(G + C^T F C)\mathbf{U})$

- Constraints from Laue condition

$k'_i = k_i$, $\vec{k}'_i = \vec{k}_i$, $\Delta k'_{i//} = \Delta k_{i//}$

$\vec{k}_i - \vec{k}'_i = \vec{G}_M$

$\Delta\vec{k}_i - \Delta\vec{k}'_i = \Delta\vec{G}_M = \begin{pmatrix} \gamma_M G_M \\ 0 \\ \Sigma_M G_M \end{pmatrix}$



$R(\theta) = \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$\Delta\vec{k}'_i = R(\theta) \begin{pmatrix} \Delta k'_{i//} \\ \Delta k'_{i\perp} \\ \Delta k'_{iz} \end{pmatrix} = \begin{pmatrix} \Delta k'_{i//}\cos\theta_M + \Delta k'_{i\perp}\sin\theta_M \\ -\Delta k'_{i//}\sin\theta_M + \Delta k'_{i\perp}\cos\theta_M \\ \Delta k'_{iz} \end{pmatrix}$

$\Delta\vec{k}_i = R(-\theta) \begin{pmatrix} \Delta k_{i//} \\ \Delta k_{i\perp} \\ \Delta k_{iz} \end{pmatrix} = \begin{pmatrix} \Delta k_{i//}\cos\theta_M - \Delta k_{i\perp}\sin\theta_M \\ \Delta k_{i//}\sin\theta_M + \Delta k_{i\perp}\cos\theta_M \\ \Delta k_{iz} \end{pmatrix}$

$$\Rightarrow \begin{cases} \gamma_M = -(\Delta k_{i\perp} + \Delta k'_{i\perp}) / 2\bar{k}_i \\ \Sigma_M = (\Delta k_{iz} - \Delta k'_{iz}) / (2\bar{k}_i \sin\theta_M) \\ \Delta k'_{i//} = \Delta k_{i//} \\ \Delta k'_{i\perp} = \Delta k_{i\perp} + 2\Delta k_{i//}\tan\theta_M \end{cases}$$

PoPovici, Acta Cryst. (1975). A31, 507

Matrix Elements In M Inverse $M^{-1} = BA(G + C^T F C)^{-1} A^T B^T$

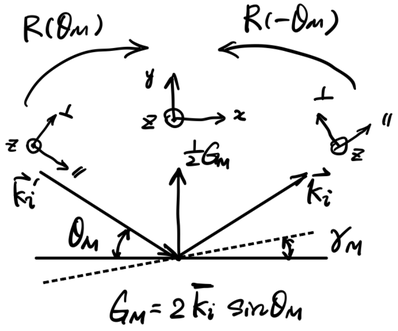
- Convert angular divergence $\mathbf{U} = (\gamma_0, \gamma_1, \delta_0, \delta_1, \gamma_2, \gamma_3, \delta_2, \delta_3)$ to uncertainty (amplitude/horizontal/vertical) in initial/final momenta k_i and k_f , $\mathbf{Y} = (\Delta k_i, k_i\gamma_1, k_i\delta_1, \Delta k_f, k_f\gamma_2, k_f\delta_2)$
- $\mathbf{Y} = A\mathbf{U}$
- $$A = \begin{bmatrix} \frac{k_i}{2\tan\theta_M} & \frac{-k_i}{2\tan\theta_M} & & & & \\ 0 & k_i & & & & \\ & & 0 & k_i & & \\ & & & & \frac{k_f}{2\tan\theta_A} & \frac{-k_f}{2\tan\theta_A} \\ & & & & k_f & 0 \\ & & & & & k_f \quad 0 \end{bmatrix}$$

- Constraints from Laue condition

$k_i' = k_i$, $\vec{k}_i' = \vec{k}_i$, $\Delta k_{i//}' = \Delta k_{i//}$

$\vec{k}_i - \vec{k}_i' = \vec{G}_M$

$\Delta\vec{k}_i - \Delta\vec{k}_i' = \Delta\vec{G}_M = \begin{pmatrix} \gamma_M G_M \\ 0 \\ \delta_M G_M \end{pmatrix}$



$R(\theta) = \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$\Delta\vec{k}_i' = R(\theta)\begin{pmatrix} \Delta k_{i//}' \\ \Delta k_{i\perp}' \\ \Delta k_{iz}' \end{pmatrix} = \begin{pmatrix} \Delta k_{i//}' \cos\theta_M + \Delta k_{i\perp}' \sin\theta_M \\ -\Delta k_{i//}' \sin\theta_M + \Delta k_{i\perp}' \cos\theta_M \\ \Delta k_{iz}' \end{pmatrix}$

$\Delta\vec{k}_i = R(-\theta)\begin{pmatrix} \Delta k_{i//} \\ \Delta k_{i\perp} \\ \Delta k_{iz} \end{pmatrix} = \begin{pmatrix} \Delta k_{i//} \cos\theta_M - \Delta k_{i\perp} \sin\theta_M \\ \Delta k_{i//} \sin\theta_M + \Delta k_{i\perp} \cos\theta_M \\ \Delta k_{iz} \end{pmatrix}$

$$\Rightarrow \begin{cases} \gamma_M = -(\Delta k_{i\perp} + \Delta k_{i\perp}')/2\bar{k}_i \\ \delta_M = (\Delta k_{iz} - \Delta k_{iz}')/(2\bar{k}_i \sin\theta_M) \\ \Delta k_{i//}' = \Delta k_{i//} \\ \Delta k_{i\perp}' = \Delta k_{i\perp} + 2\Delta k_{i//} \tan\theta_M \end{cases}$$

PoPovici, Acta Cryst. (1975). A31, 507

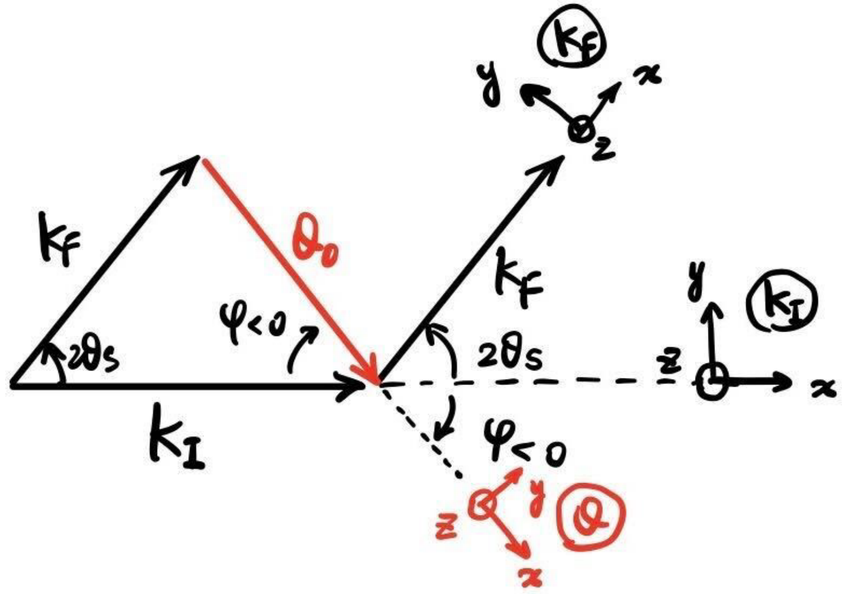
Matrix Elements In M Inverse $M^{-1} = BA(G + C^T F C)^{-1} A^T B^T$

- Convert uncertainty (amplitude/horizontal/vertical) in initial/final momenta k_i and k_f , $\mathbf{Y} = (\Delta k_i, k_i\gamma_1, k_i\delta_1, \Delta k_f, k_f\gamma_2, k_f\delta_2)$ to uncertainty in momentum and energy transfer $\mathbf{X} = (\mathbf{Q}, E)$

- $\mathbf{X} = B\mathbf{Y}$

- $$B = \begin{bmatrix} \cos\phi & \sin\phi & 0 & -\cos(\phi - 2\theta_S) & -\sin(\phi - 2\theta_S) & 0 \\ -\sin\phi & \cos\phi & 0 & \sin(\phi - 2\theta_S) & -\cos(\phi - 2\theta_S) & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 \\ \frac{\hbar^2}{m} k_i & 0 & 0 & -\frac{\hbar^2}{m} k_f & 0 & 0 \end{bmatrix}$$, where

$$\phi = \arctan \frac{-k_f \sin(2\theta_S)}{k_i - k_f \cos(2\theta_S)} . (-\pi < \phi \leq \pi)$$

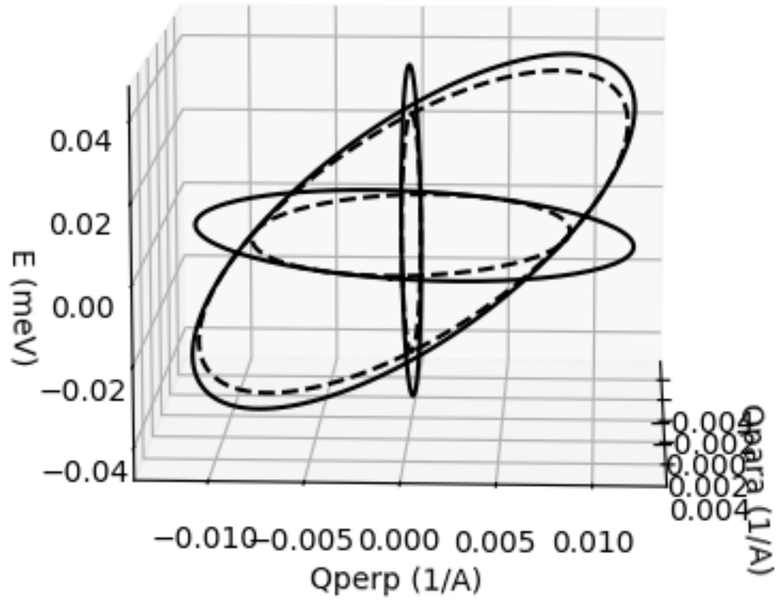


- Popovici method: $M^{-1} = BA\left(G + [D(S + T^T F T)D^T]^{-1}\right)^{-1} A^T B^T$

PoPovici, Acta Cryst. (1975). A31, 507

Contents

- A measurement by the Triple-Axis Spectrometer
  - Intensity
  - Broadening from instrument
- Resolution function from Cooper-Nathans's method
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  - Diffraction: deconvolution for correlation length
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Quadratic Form of an Ellipsoid

- Resolution function

$$R(\mathbf{Q}, E) = R_0 \cdot \frac{\sqrt{|M|}}{4\pi^2} \exp\left(-\frac{1}{2}\begin{bmatrix}\mathbf{Q}\\ E\end{bmatrix}^T M \begin{bmatrix}\mathbf{Q}\\ E\end{bmatrix}\right)$$

- Resolution matrix M can be described by a 4D ellipsoid

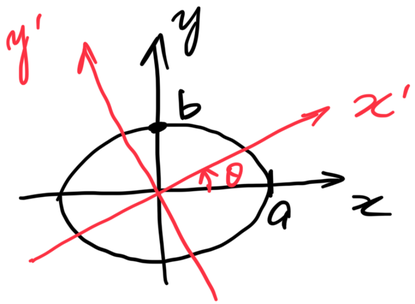
Ellipses can be described as Quadratics

$$\begin{pmatrix}x\\ y\end{pmatrix}^T \underbrace{\begin{pmatrix}\frac{1}{a^2} & 0\\ 0 & \frac{1}{b^2}\end{pmatrix}} \begin{pmatrix}x\\ y\end{pmatrix} = \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$\varepsilon_1 = \frac{1}{a^2}$ $v_1 = \begin{pmatrix}1\\ 0\end{pmatrix}$, $\frac{1}{\sqrt{\varepsilon_1}} = a$

$\varepsilon_2 = \frac{1}{b^2}$ $v_2 = \begin{pmatrix}0\\ 1\end{pmatrix}$ $\frac{1}{\sqrt{\varepsilon_2}} = b$

Half-Axes



Quadratic Form of an Ellipsoid

- Resolution function

$$R(\mathbf{Q}, E) = R_0 \cdot \frac{\sqrt{|M|}}{4\pi^2} \exp\left(-\frac{1}{2}\begin{bmatrix}\mathbf{Q}\\ E\end{bmatrix}^T M \begin{bmatrix}\mathbf{Q}\\ E\end{bmatrix}\right)$$

- Resolution matrix M can be described by a 4D ellipsoid
- Half-axes of the ellipsoid are $\frac{1}{\sqrt{\lambda_i}}$, where λ_i are ***eigenvalues of M*** (Notice the annoyance FWHM $\Gamma = 2\sqrt{\ln 2}\sigma$)
- Principal axes of the ellipsoid are ***eigenvectors of M***

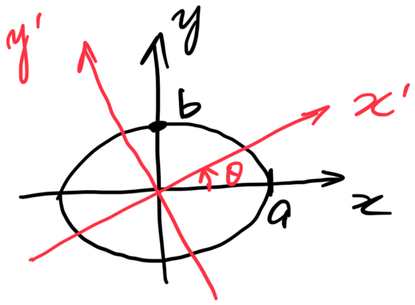
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Half-Axes



Quadratic Form of an Ellipse

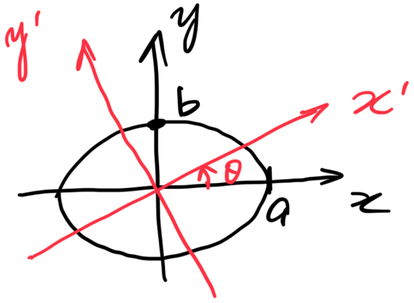
- Resolution function

$$R(\mathbf{Q}, E) = R_0 \cdot \frac{\sqrt{|M|}}{4\pi^2} \exp\left(-\frac{1}{2}\begin{bmatrix}\mathbf{Q}\\ E\end{bmatrix}^T M \begin{bmatrix}\mathbf{Q}\\ E\end{bmatrix}\right)$$

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- Principal axes of the ellipsoid are ***eigenvectors of M***

If having a θ ccw Rotated axes

$$\begin{pmatrix} x \\ y \end{pmatrix} = \underbrace{\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}}_{\text{2D Rotation Matrix}} \begin{pmatrix} x' \\ y' \end{pmatrix}$$

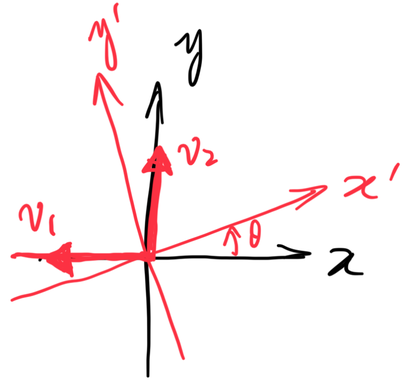


$$1 = \begin{pmatrix} x' \\ y' \end{pmatrix}^T \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} \frac{1}{a^2} & 0 \\ 0 & \frac{1}{b^2} \end{pmatrix} \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x' \\ y' \end{pmatrix}$$

$$= \begin{pmatrix} x' \\ y' \end{pmatrix}^T \begin{pmatrix} \frac{1}{a^2}\cos^2\phi + \frac{1}{b^2}\sin^2\phi & \left(\frac{-1}{a^2}+\frac{1}{b^2}\right)\cos\phi\sin\phi \\ \left(\frac{-1}{a^2}+\frac{1}{b^2}\right)\cos\phi\sin\phi & \frac{1}{b^2}\cos^2\phi + \frac{1}{a^2}\sin^2\phi \end{pmatrix} \begin{pmatrix} x' \\ y' \end{pmatrix}$$

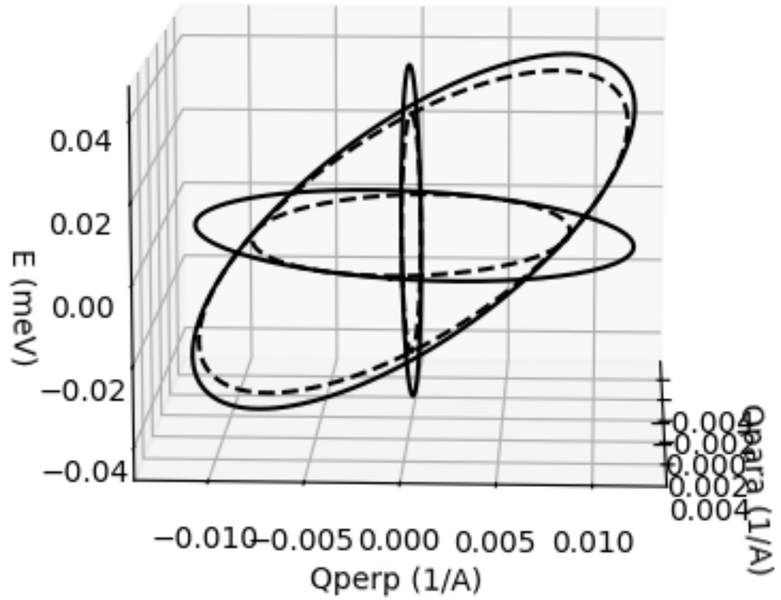
$$\varepsilon_1 = \frac{1}{a^2} \qquad v_1 = \begin{pmatrix} -\cos\theta \\ \sin\theta \end{pmatrix}$$

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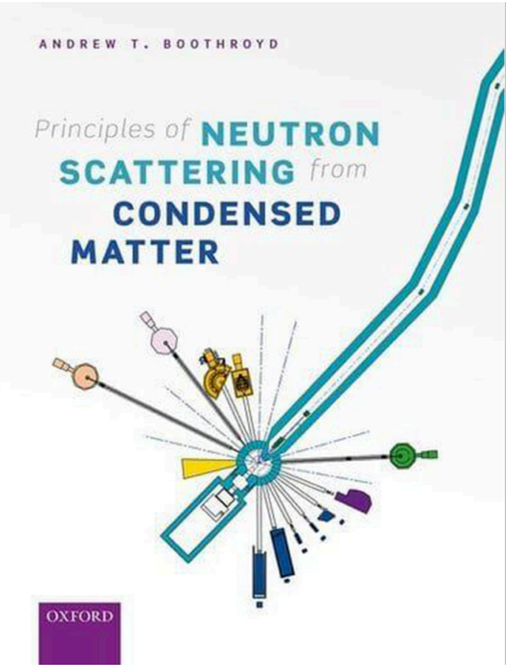


Contents

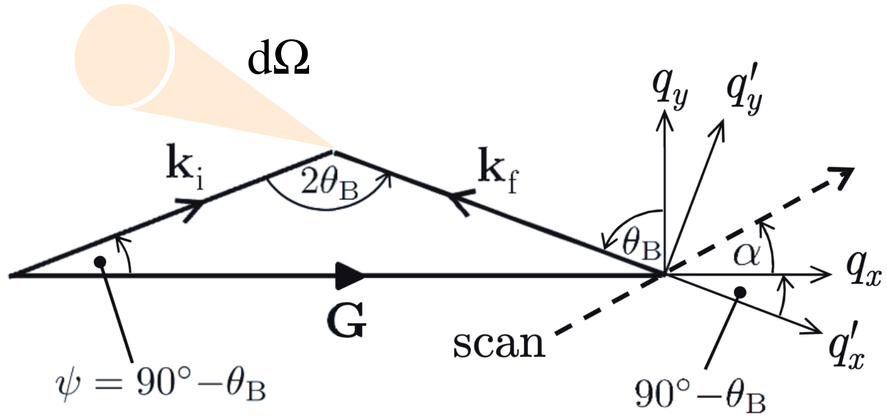
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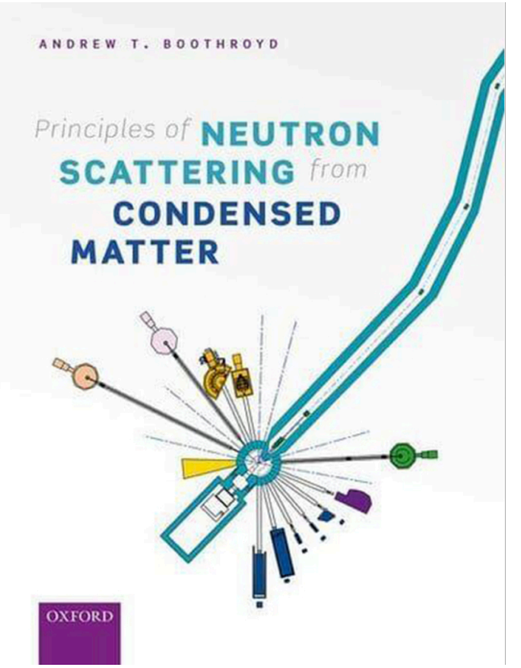
Integrated Intensity and Lorentz Factor (Open Detector)



- Differential cross section $\left(\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right)(q'_x, q'_y, q'_z) = N\frac{(2\pi)^3}{v_0}\sum_{\mathbf{G}} |F(\mathbf{G})|^2 \delta(\mathbf{Q}-\mathbf{G})$
- Measured count rate $\sigma(q'_x) = \Phi_{in}\int\left(\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right)\mathrm{d}\Omega$, where $\mathrm{d}\Omega = \frac{\mathrm{d}k'_{\mathrm{fy}}\mathrm{d}k'_{\mathrm{fz}}}{k_{\mathrm{f}}^2} = \frac{\mathrm{d}q'_y\mathrm{d}q'_z}{k^2}$
- Integrated intensity ***along path-s*** $I_s(\mathbf{G}) = \int\sigma\left(q'_x\right)\mathrm{d}s$



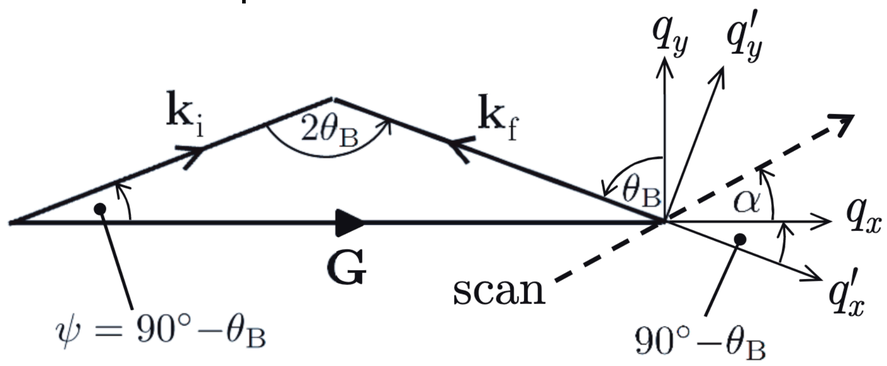
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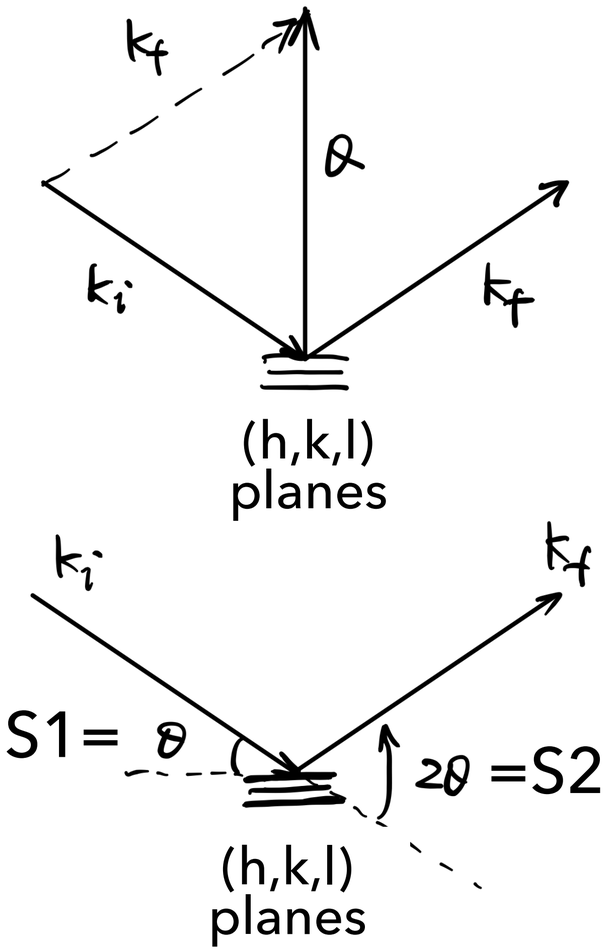
$$I_s(\mathbf{G}) = \int \sigma\left(q'_x\right)\mathrm{d}s = \frac{1}{\left|\sin\left(\theta_{\mathrm{B}}-\alpha\right)\right|}\int \sigma\left(q'_x\right)\mathrm{d}q'_x = \Phi N\frac{(2\pi)^3}{v_0}|F(\mathbf{G})|^2\frac{1}{k^2}\frac{1}{\left|\sin\left(\theta_{\mathrm{B}}-\alpha\right)\right|}$$

- $$= \Phi\frac{2\pi V}{v_0^2}\lambda^2|F(\mathbf{G})|^2\frac{1}{\left|\sin\left(\theta_{\mathrm{B}}-\alpha\right)\right|}$$
- Lorentz factor $L = \frac{1}{\left|\sin\left(\theta_{\mathrm{B}}-\alpha\right)\right|}$



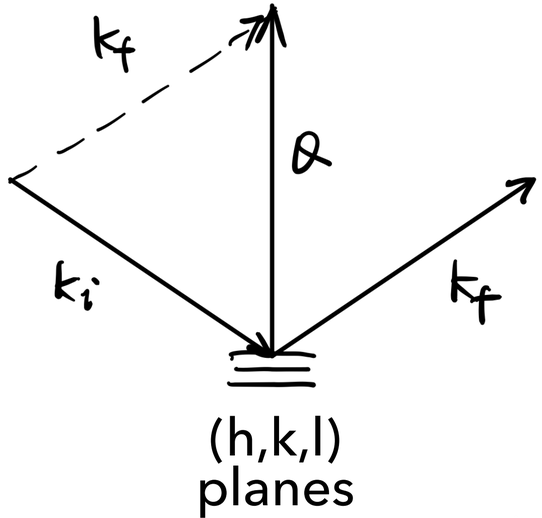
S1-Scans and Th2Th Scans

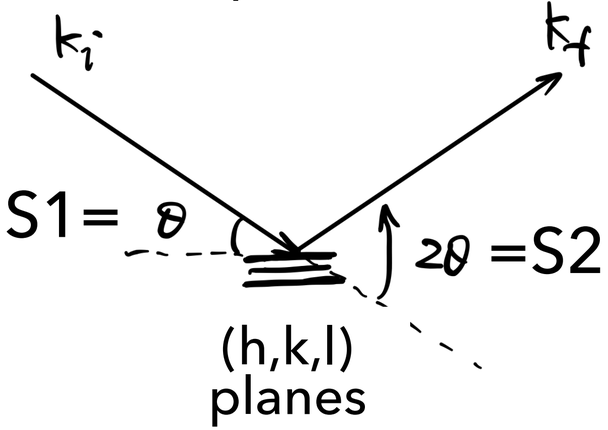
- $2d_{(h,k,l)} \sin\theta = \lambda$
- S1-scan
  - Changes s1 only
  - Length of **Q** fixed
  - Transverse scan
  - Probes sample mosaic
- Th2th scan
  - Changes s1 and s2
  - Direction of **Q** fixed
  - Longitudinal scan
  - Probes lattice parameters

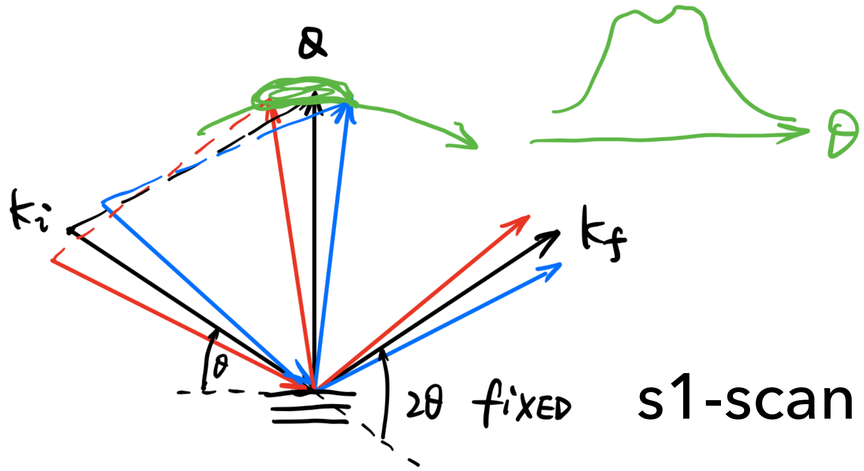


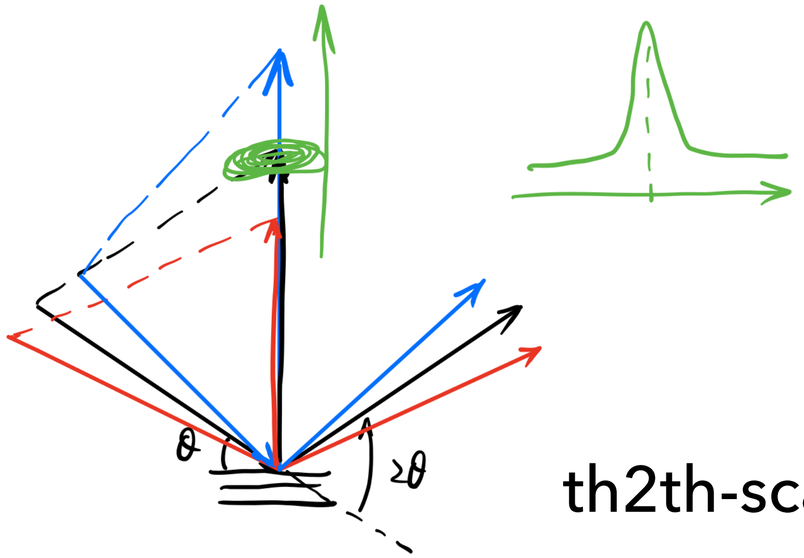
S1-Scans and Th2Th Scans

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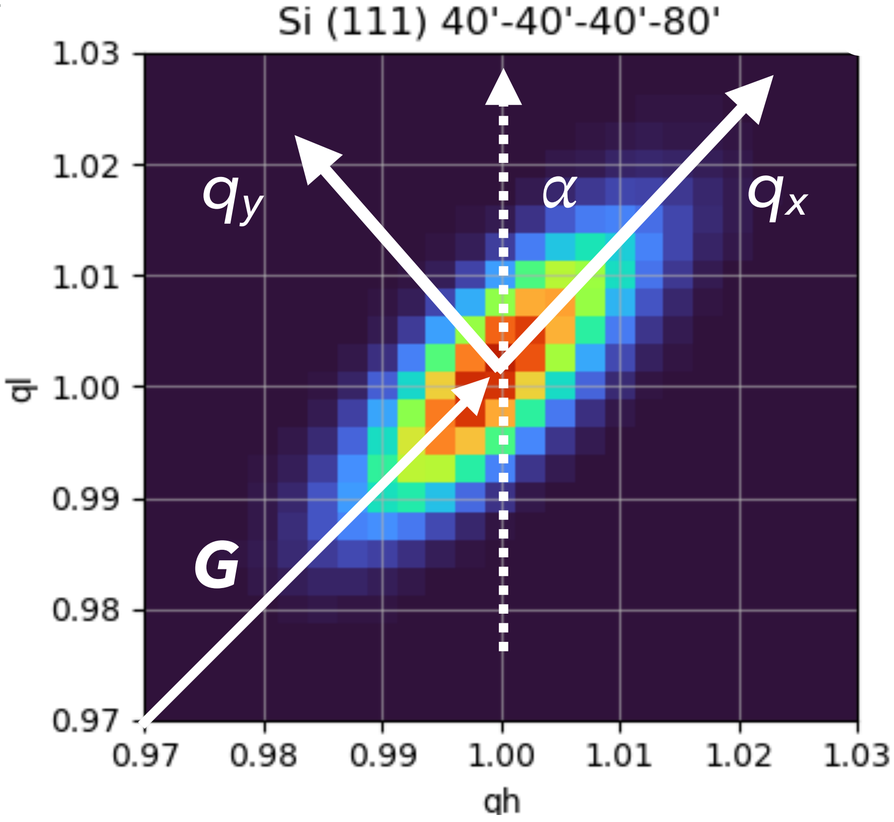


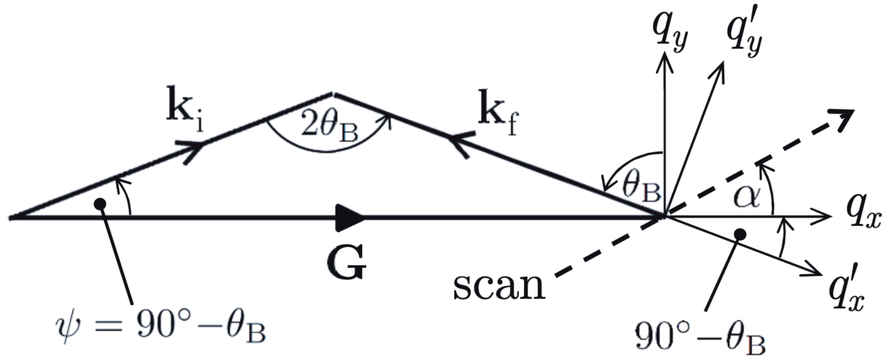




Lorentz Factor for S1-Scans and Th2Th-Scans

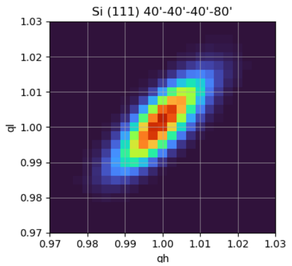
- $I_s(\mathbf{G}) = \int \Sigma\left(q_x'\right) \mathrm{d}s = \Phi \frac{2\pi V}{v_0^2} \lambda^2 |F(\mathbf{G})|^2 \frac{1}{\left|\sin\left(\theta_\mathrm{B} - \alpha\right)\right|}$
- Th2th-scans:
  - α=0, $L = \frac{1}{\sin\theta_B}$
  - $\mathrm{d}s = 2k\cos\theta_B \mathrm{d}\omega$, where $\mathrm{d}\omega = \frac{\mathrm{d}q_x'}{k\sin 2\theta_B}$
  - $I_{\omega-\theta}(\mathbf{G}) = \Phi \frac{V}{v_0^2} \lambda^3 |F(\mathbf{G})|^2 \frac{1}{\sin 2\theta_\mathrm{B}}$
- S1-scans:
  - α=90 deg, $L = \frac{1}{\cos\theta_B}$
  - $\mathrm{d}s = G\mathrm{d}\omega = 2k\sin\theta_B \mathrm{d}\omega$, where $\mathrm{d}\omega = \frac{\mathrm{d}q_x'}{k\sin 2\theta_B}$
  - $I_{\omega}(\mathbf{G}) = \Phi \frac{V}{v_0^2} \lambda^3 |F(\mathbf{G})|^2 \frac{1}{\sin 2\theta_\mathrm{B}}$
- **With Open Detector, Lorentz factor or both LG and TR syncs are** $\frac{1}{\sin 2\theta_B}$





Integrated Intensity and Lorentz Factor (Triple-Axis Spectrometer w/ Analyzer)

- Differential cross section $\left(\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right)_{\mathrm{eff}} = N\frac{(2\pi)^3}{v_0}\sum_{\mathbf{G}} |F(\mathbf{G})|^2 \int \delta(\mathbf{Q}-\mathbf{G})R(\mathbf{Q_0}-\mathbf{Q})\,\mathrm{d}^3\mathbf{Q}$

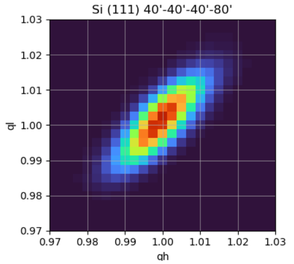


- Resolution function

$$R(\mathbf{Q}_0-\mathbf{Q}) = \frac{\sqrt{|M|}}{2\pi}\exp\left(-\frac{1}{2}M_{11}q_x^2 - M_{12}q_xq_y - \frac{1}{2}M_{22}q_y^2\right) = \frac{\sqrt{|M|}}{2\pi}\exp\left(-\frac{1}{2}\begin{bmatrix}q_x\\q_y\end{bmatrix}^T\begin{bmatrix}M_{11} & M_{12}\\M_{12} & M_{22}\end{bmatrix}\begin{bmatrix}q_x\\q_y\end{bmatrix}\right)$$

Integrated Intensity and Lorentz Factor (Triple-Axis Spectrometer w/ Analyzer)

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- Resolution function

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- Integrated intensity ***along path-s***

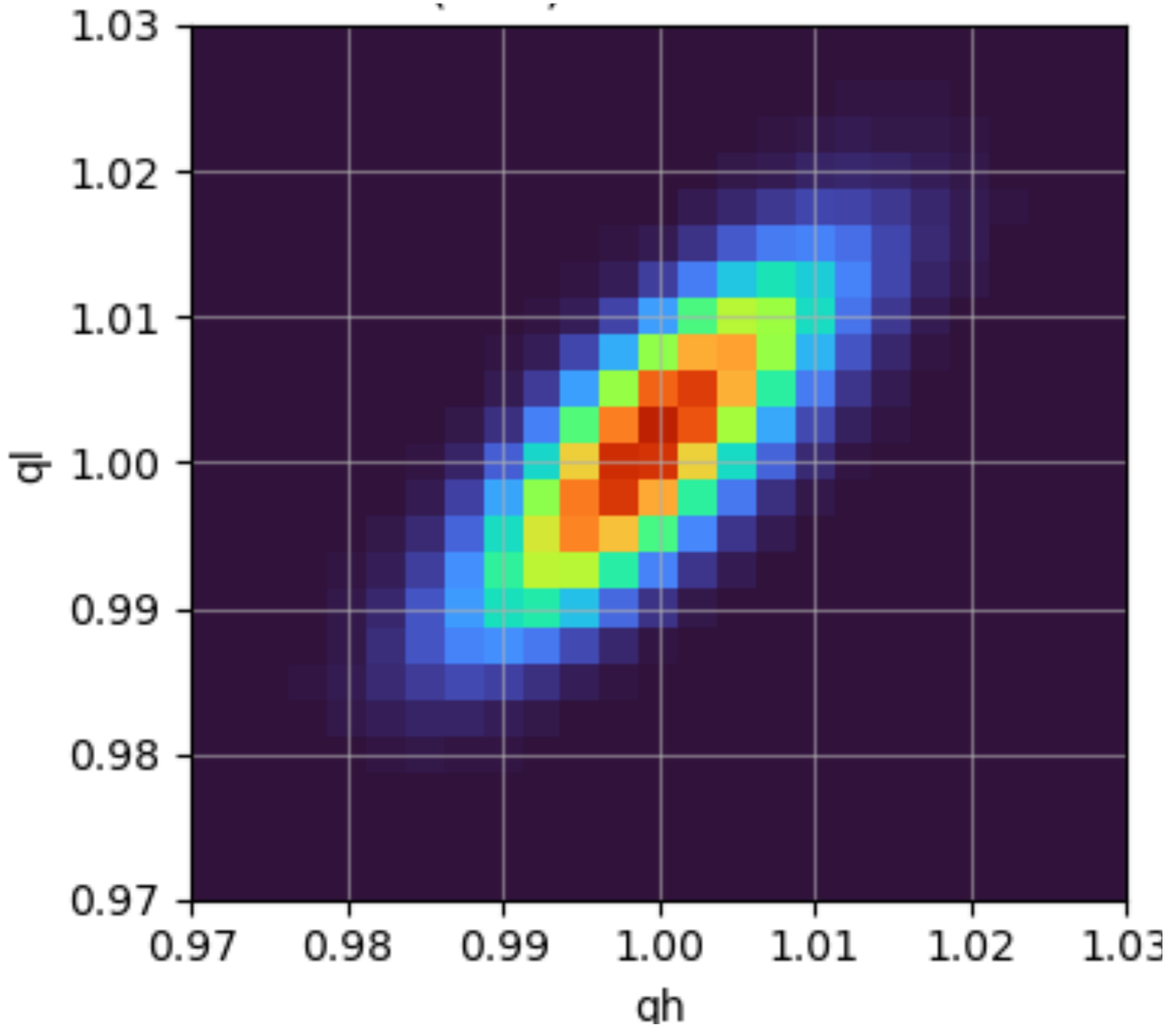
$$I_s(\mathbf{G}) = \int\Sigma\left(q_x'\right)\mathrm{d}s = C|F(\mathbf{G})|^2\int R(\mathbf{Q_0}-\mathbf{G})\,\mathrm{d}s$$

$$= \frac{C|F(\mathbf{G})|^2}{(2\pi)^{1/2}}\frac{|\mathrm{M}|^{1/2}}{\left(M_{11}\cos^2\alpha + 2M_{12}\cos\alpha\sin\alpha + M_{22}\sin^2\alpha\right)^{1/2}}$$

- Lorentz factor $L = \dfrac{|\mathrm{M}|^{1/2}}{\left(M_{11}\cos^2\alpha + 2M_{12}\cos\alpha\sin\alpha + M_{22}\sin^2\alpha\right)^{1/2}}$

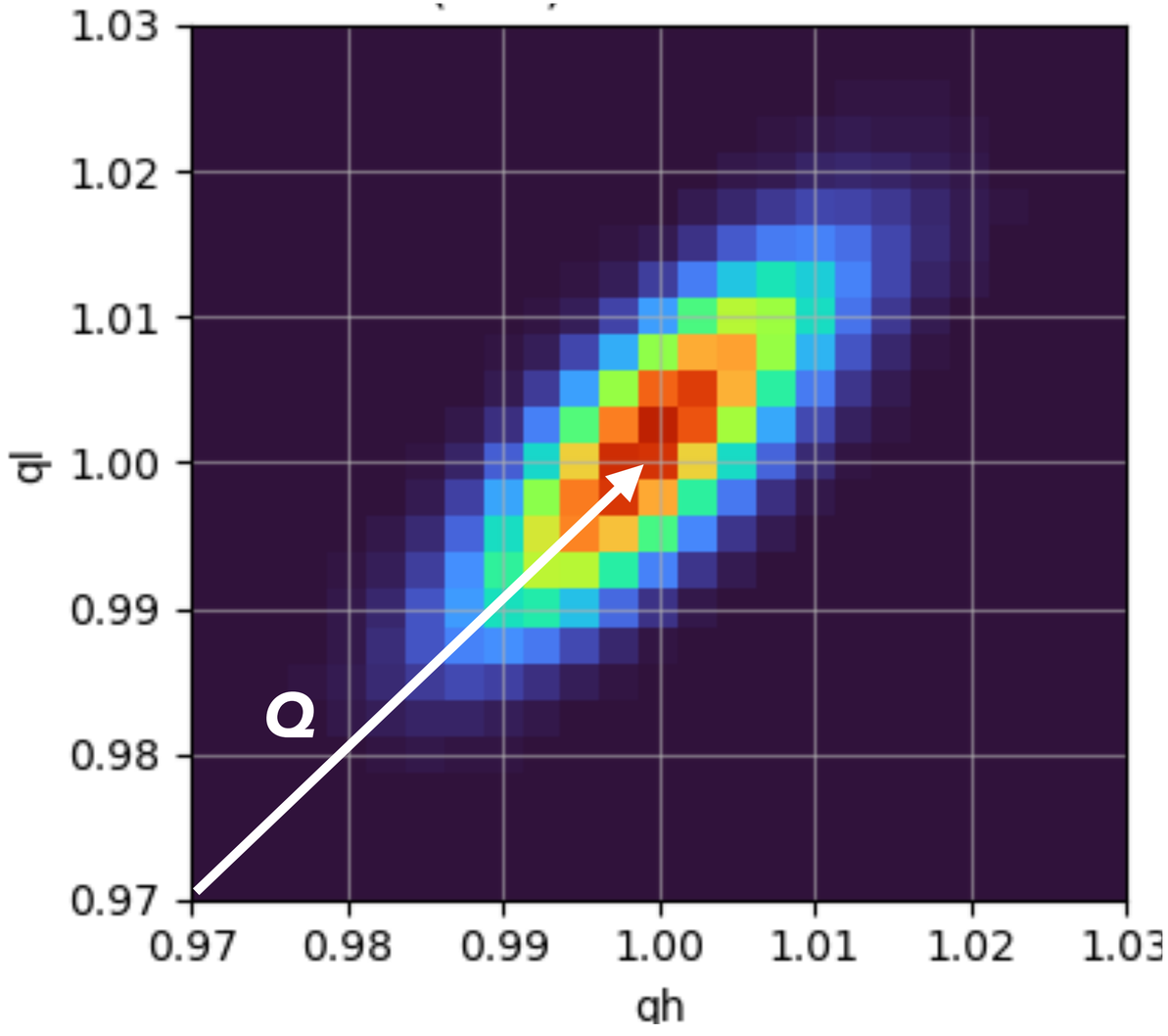
Intuition About Removal of the Lorentz Factor

- Gaussian $g(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp(-\frac{x^2}{2\sigma^2})$,
  - FWHM=$2\sqrt{2\ln 2}\sigma$,
  - height=$\frac{1}{\sigma\sqrt{(2\pi)}}$,
  - area = 1 ~ height * FWHM



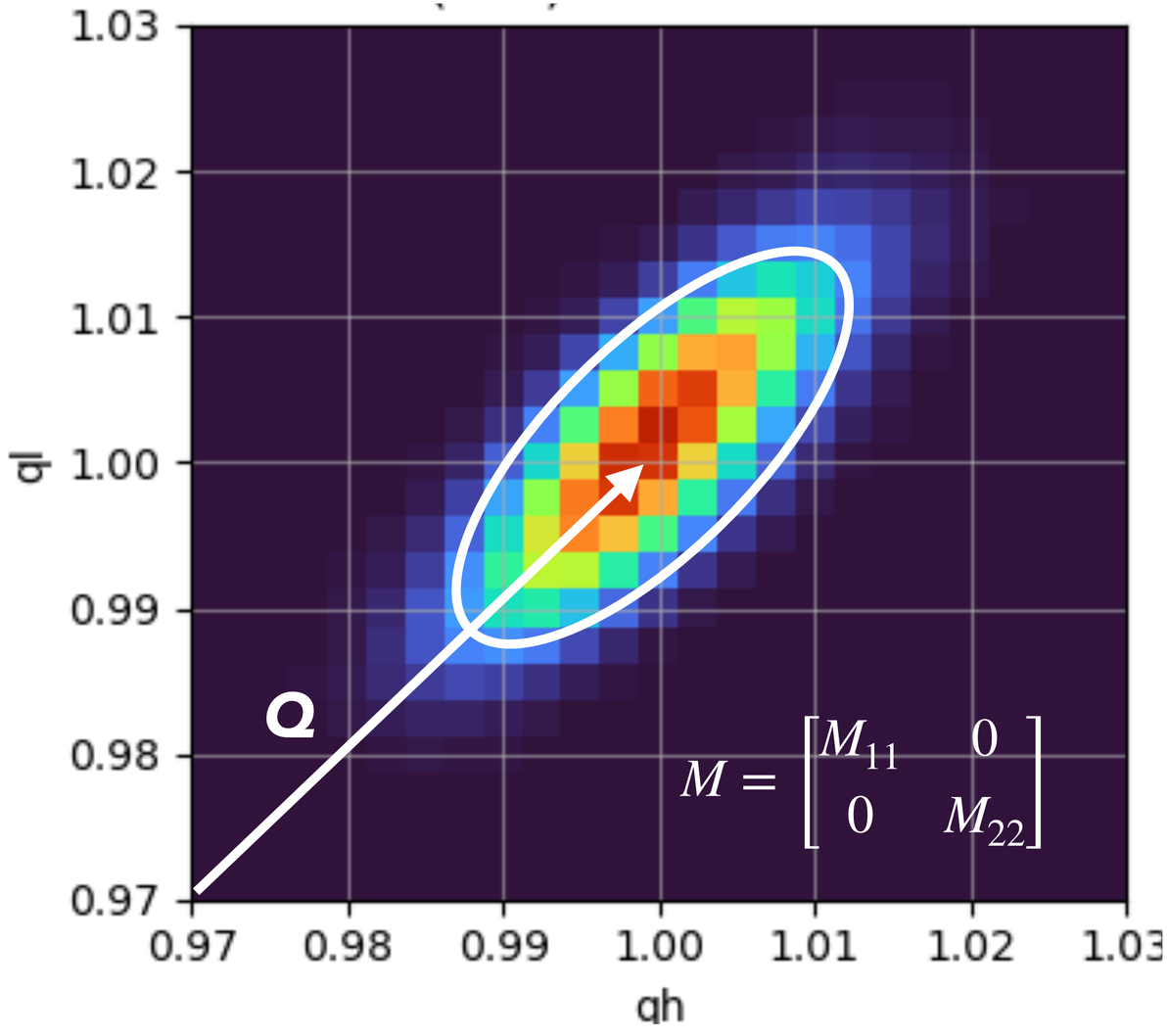
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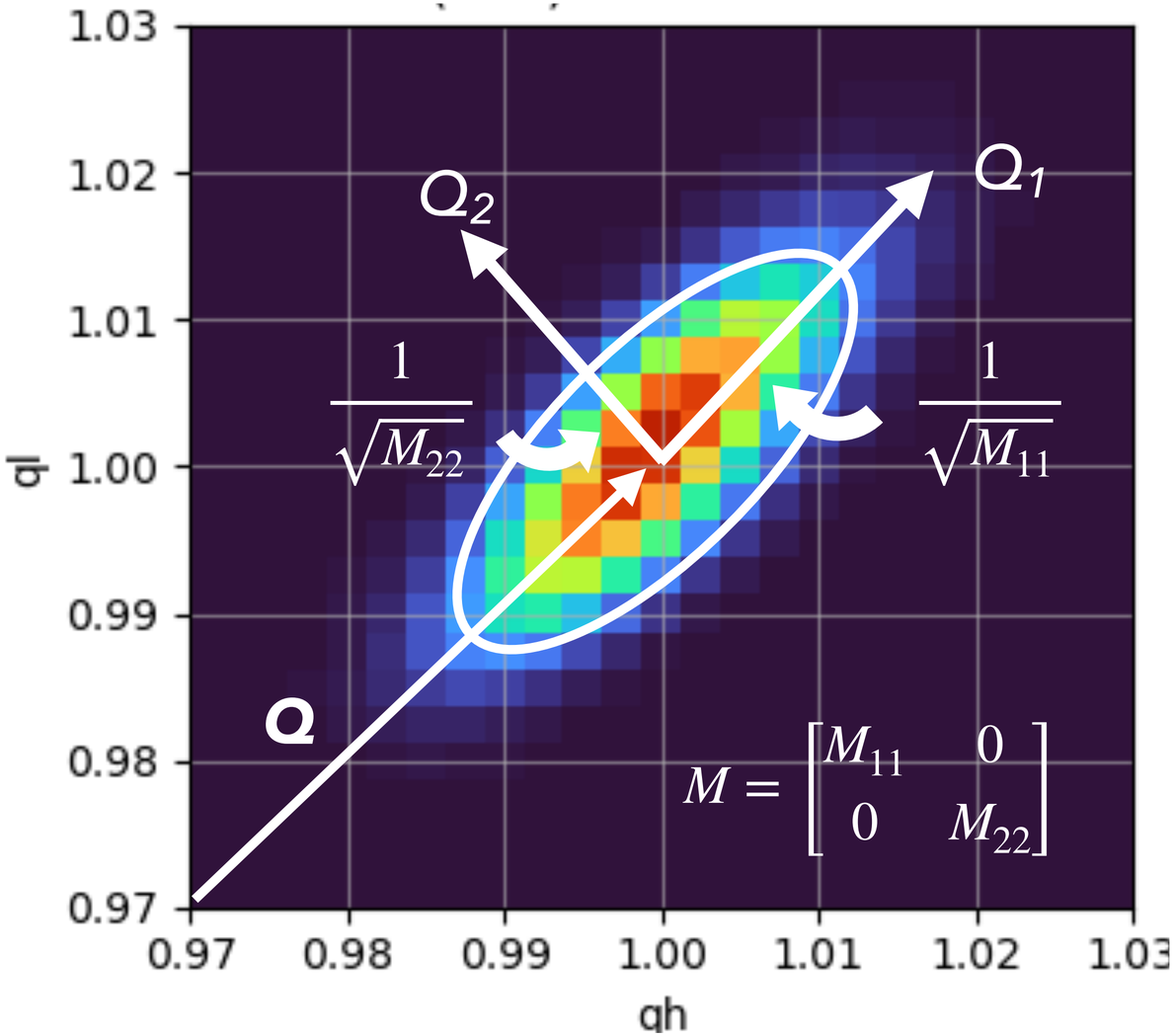
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Intuition About Removal of the Lorentz Factor

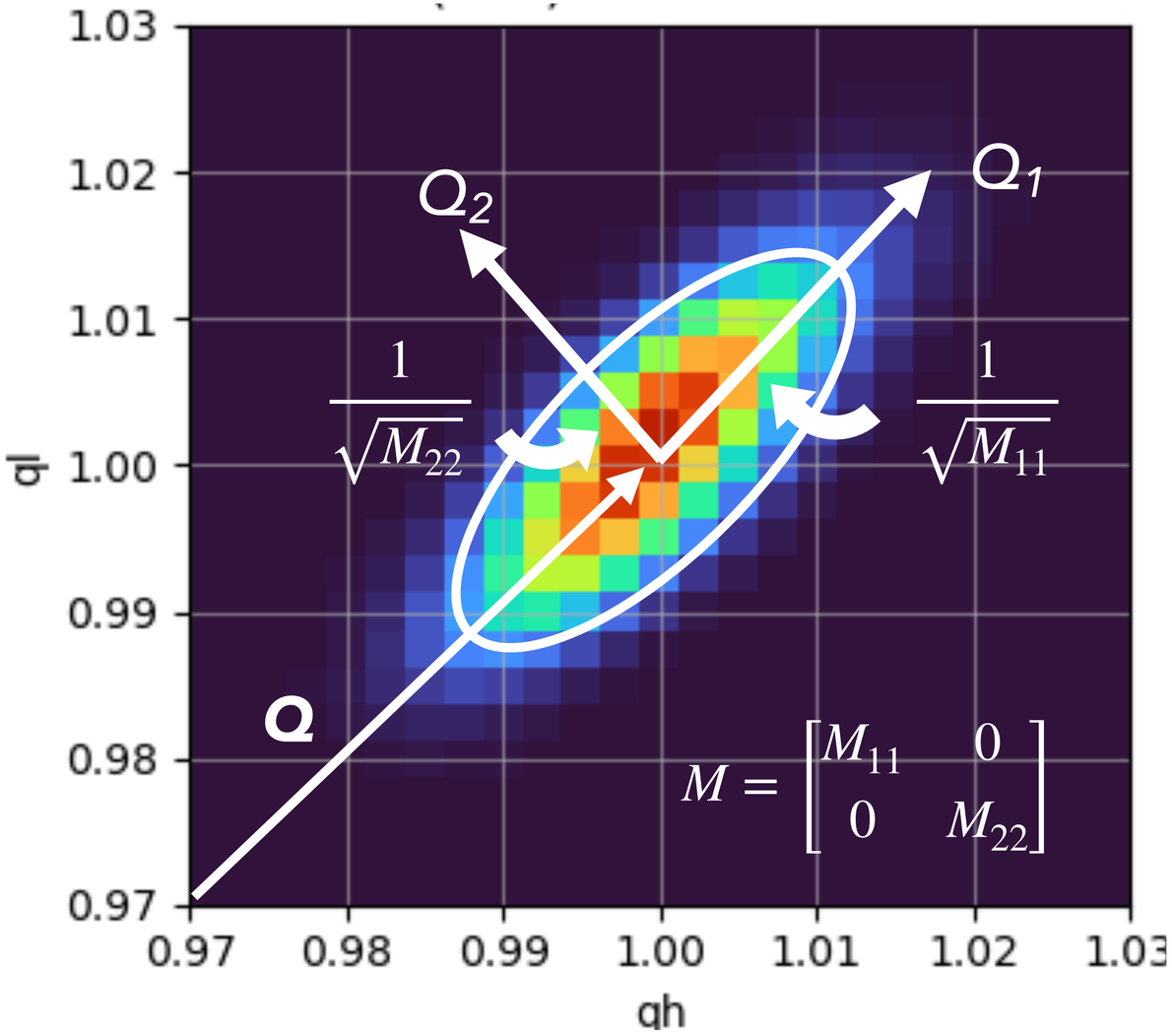
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  - FWHM=$2\sqrt{2\ln 2}\sigma$,
  - height=$\frac{1}{\sigma\sqrt{(2\pi)}}$,
  - area = 1 ~ height * FWHM

- $|F|^2 \sim \frac{I_{\text{int}}}{L}$

- LG scan: α=0, $L = \frac{1}{\sqrt{M_{11}}} \cdot \frac{\sqrt{|M|}}{2\cos\theta}$

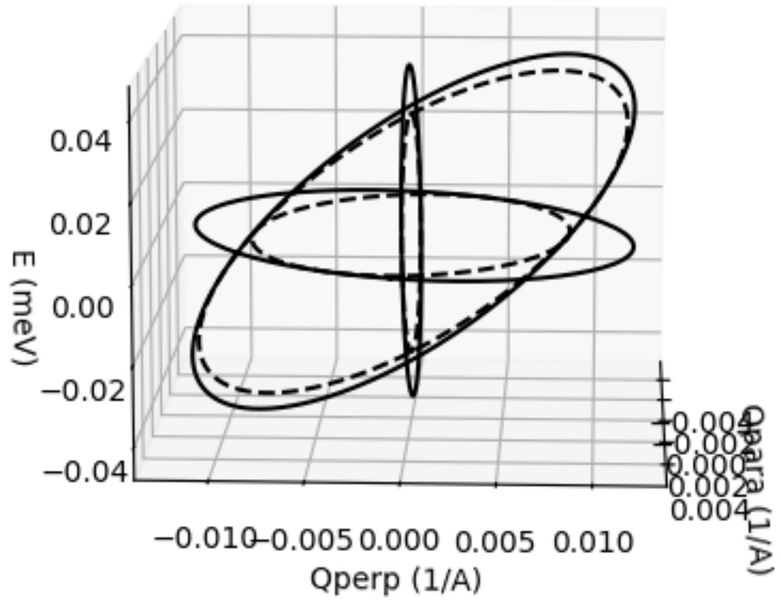
- TR scan: α=90 deg, $L = \frac{1}{\sqrt{M_{22}}} \cdot \frac{\sqrt{|M|}}{2|\sin\theta|}$

- Removal of the Lorentz factor gives the Height of the peak



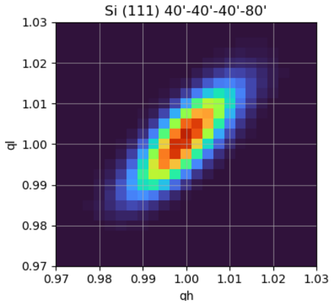
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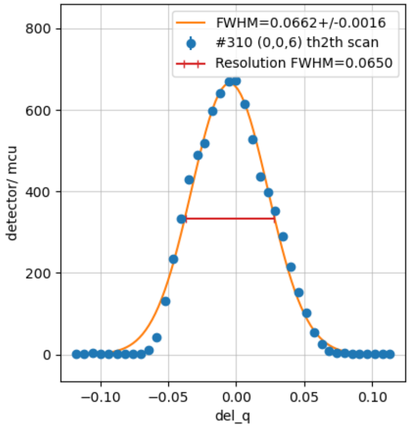


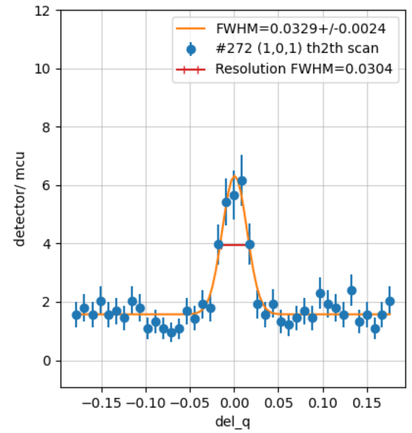
Workflow for Refinement of Nuclear/Magnetic Structure

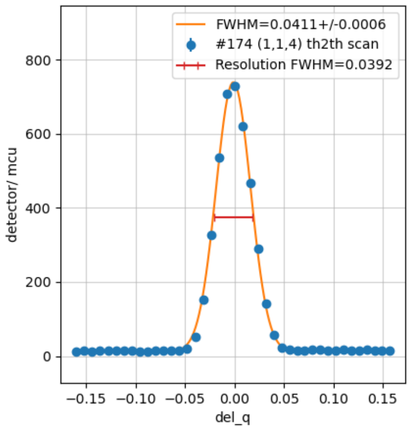
- Fit LG and TR scans. Exclude ones having mismatched background/ height



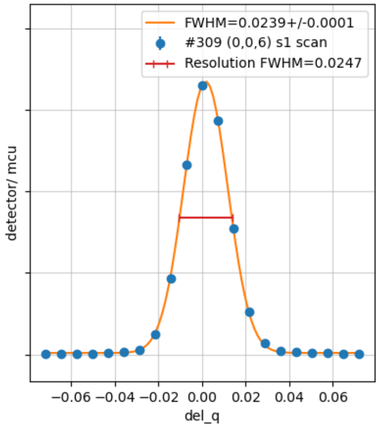
LG

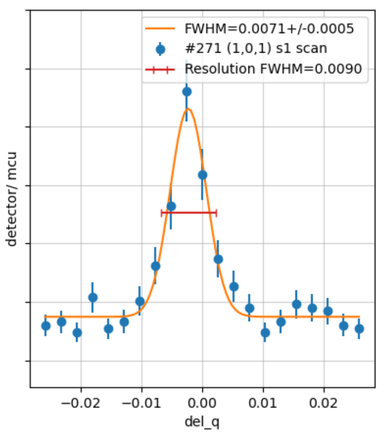


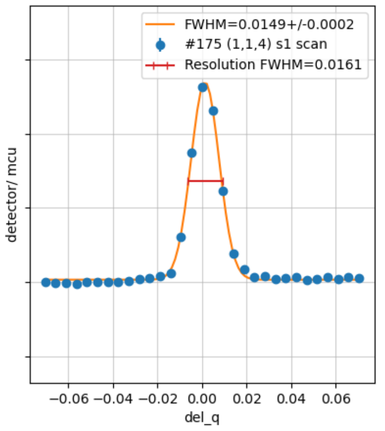




TR

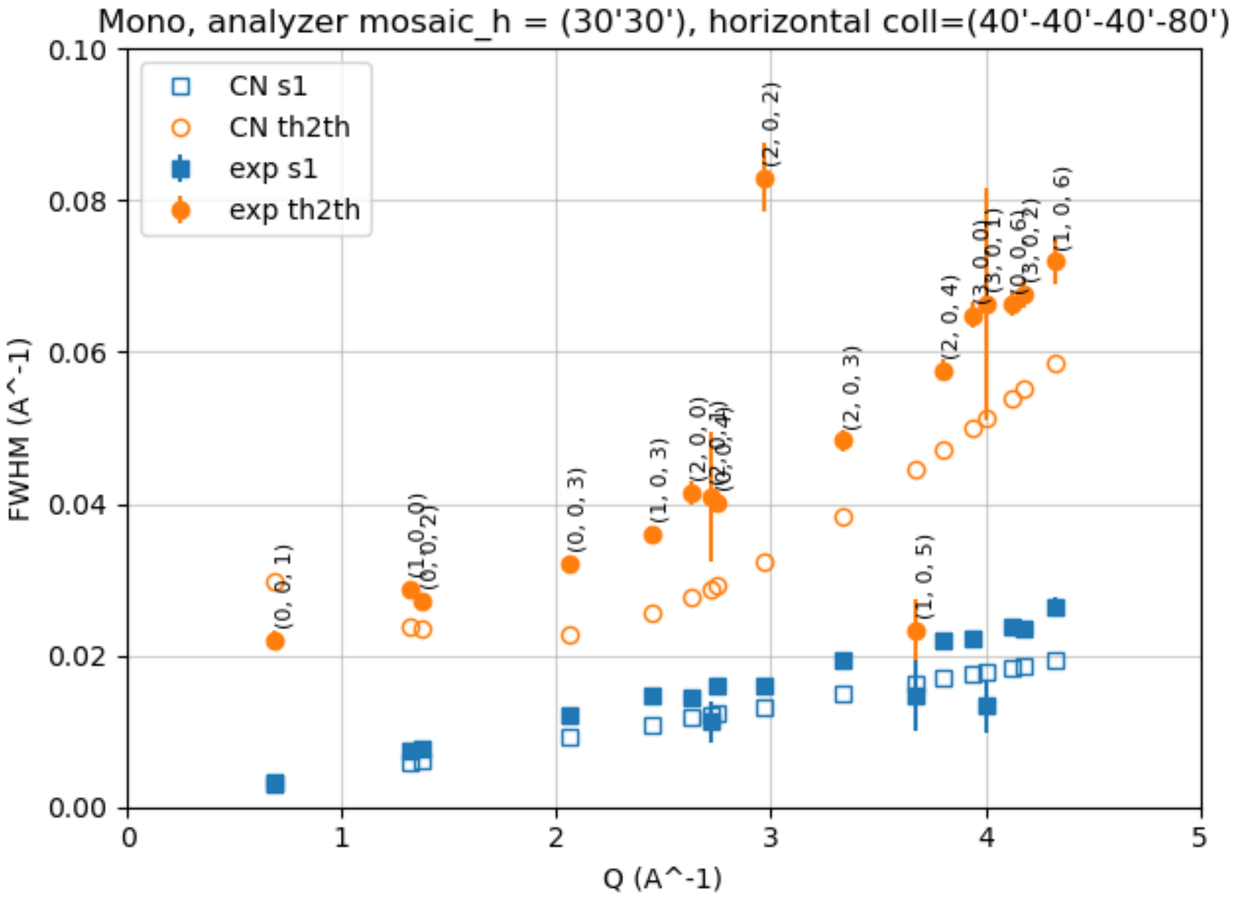






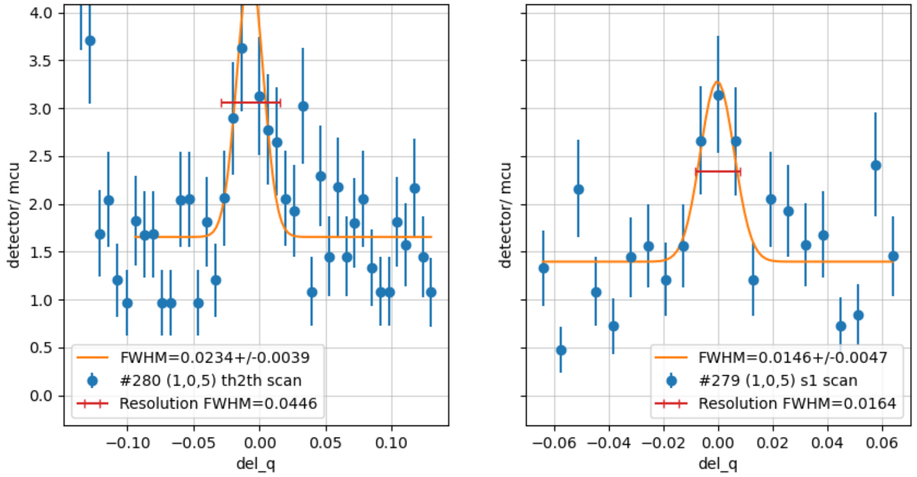
Workflow for Refinement of Nuclear/Magnetic Structure

- Fit LG and TR scans. Exclude ones having mismatched background/ height
- Plot FWHM vs. Q. Exclude bad peaks.
- Tune instrument/sample parameters

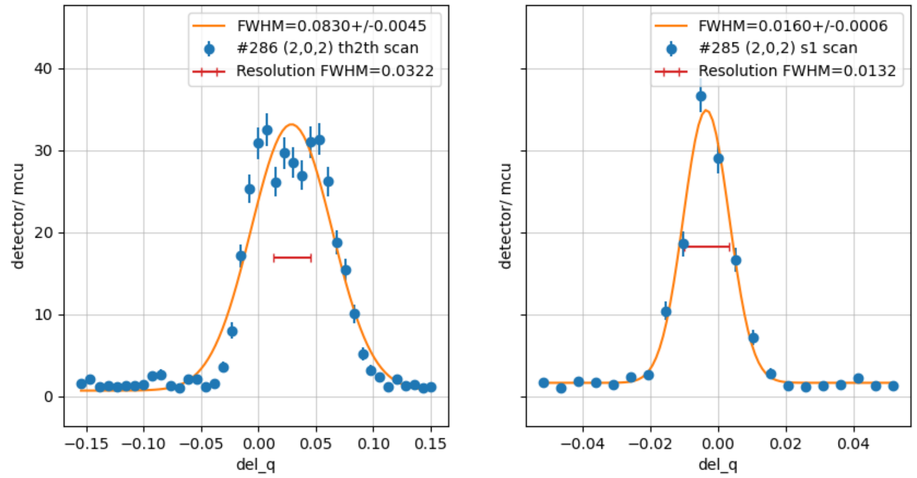


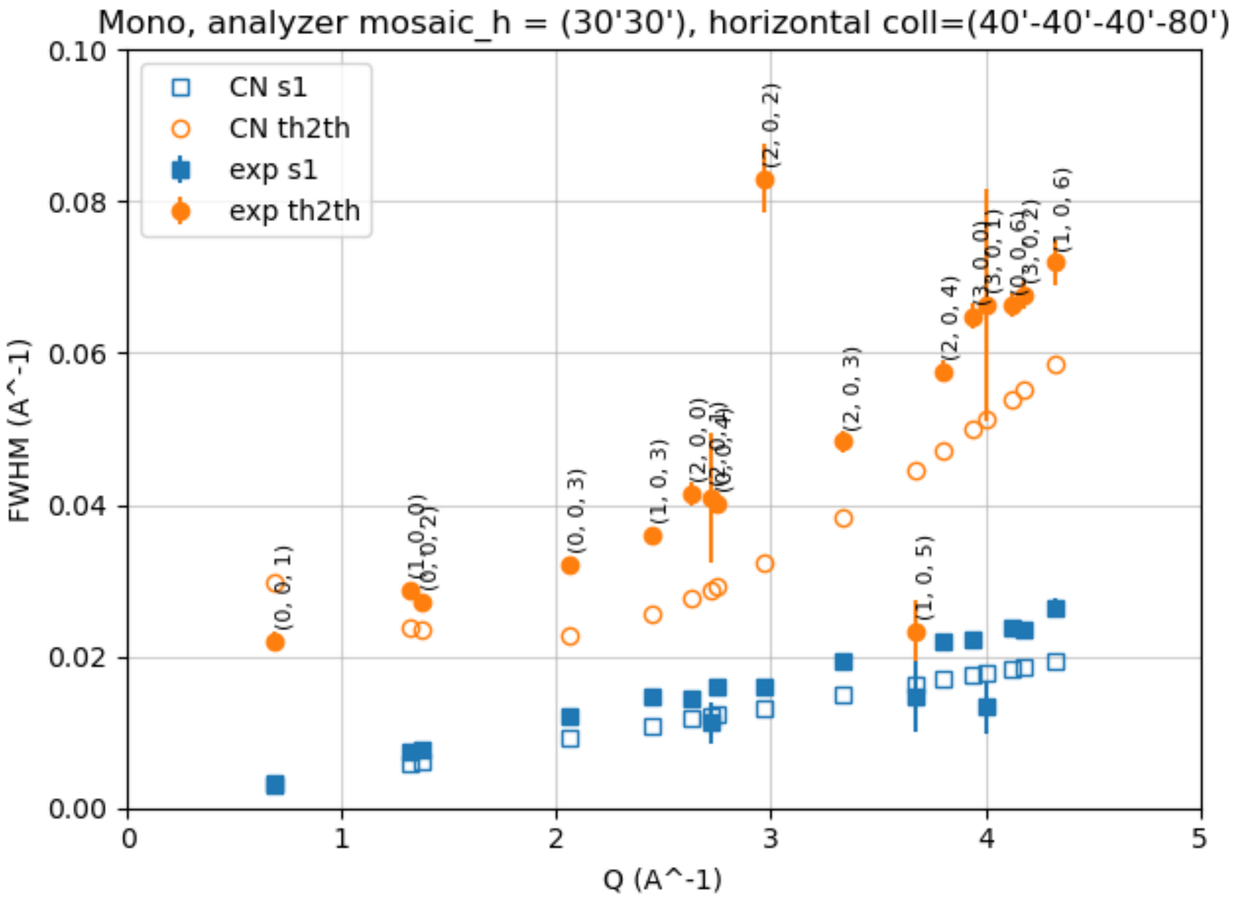
Workflow for Refinement of Nuclear/Magnetic Structure

(1, 0, 5)



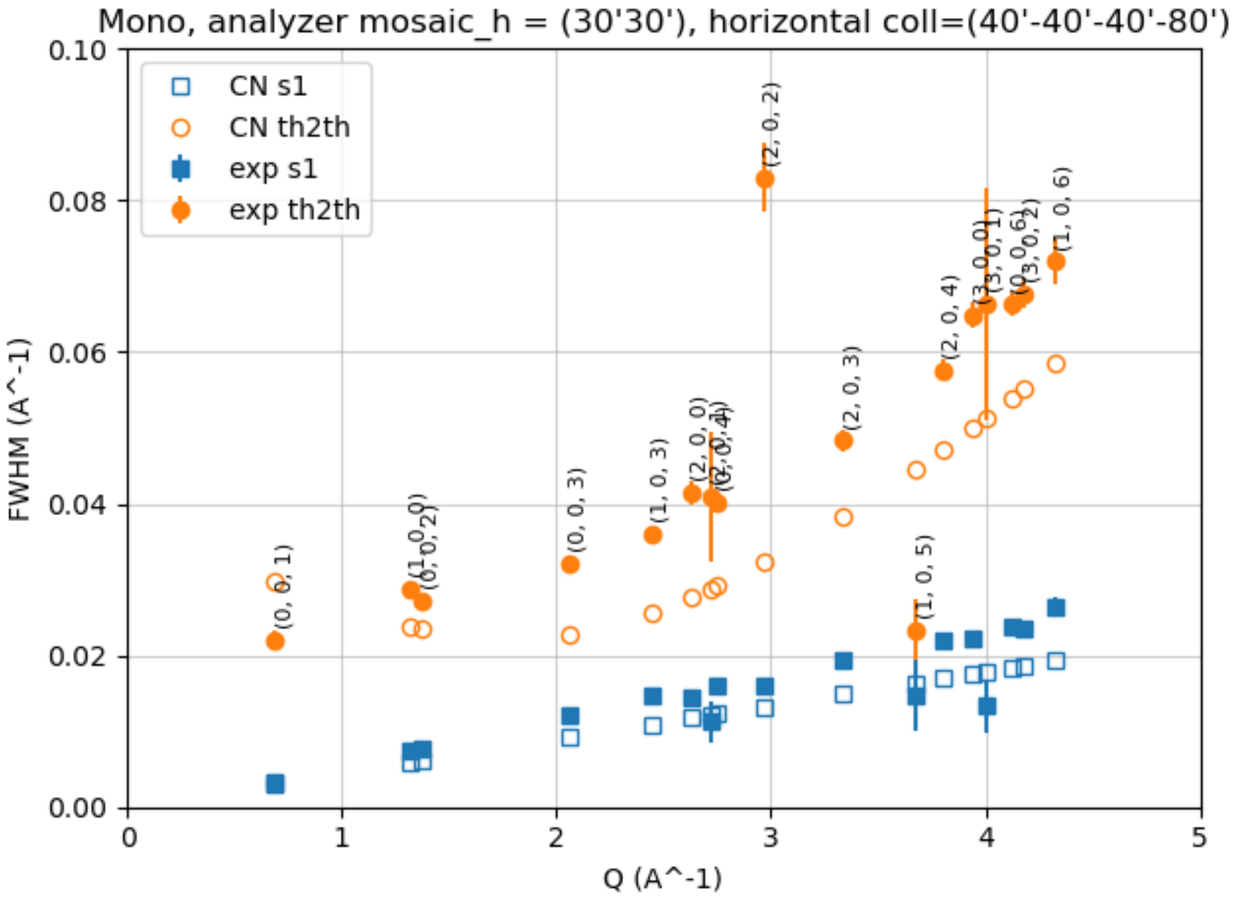
(2, 0, 2)





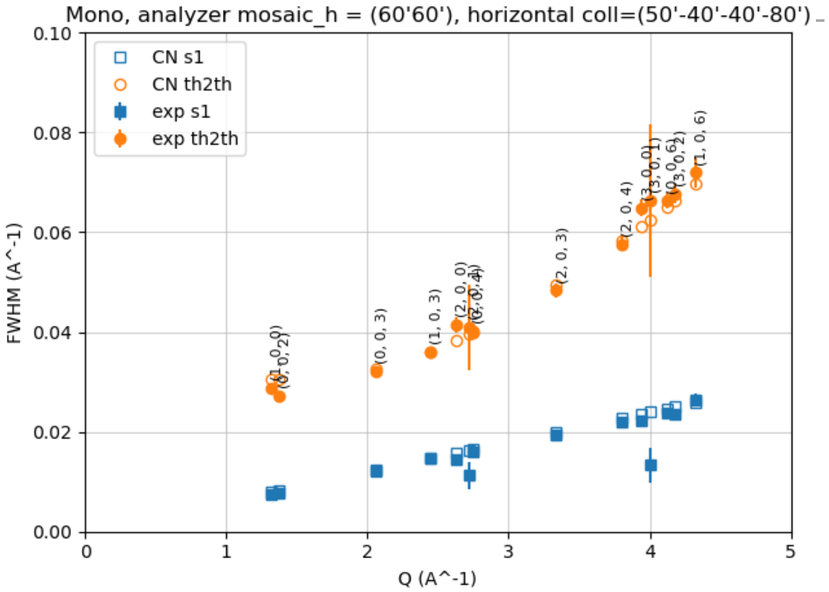
Workflow for Refinement of Nuclear/Magnetic Structure

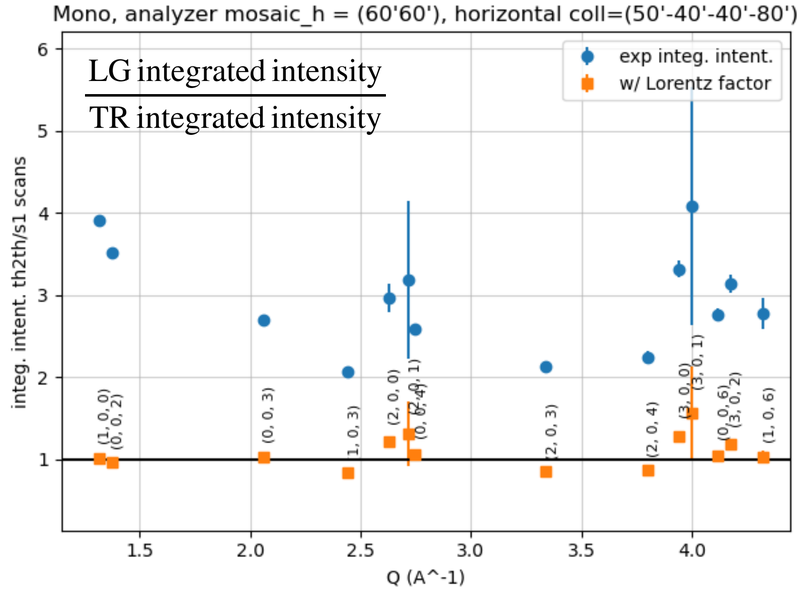
- Fit LG and TR scans. Exclude ones having mismatched background/ height
- Plot FWHM vs. Q. Exclude bad peaks.
- Tune instrument/sample parameters



Workflow for Refinement of Nuclear/Magnetic Structure

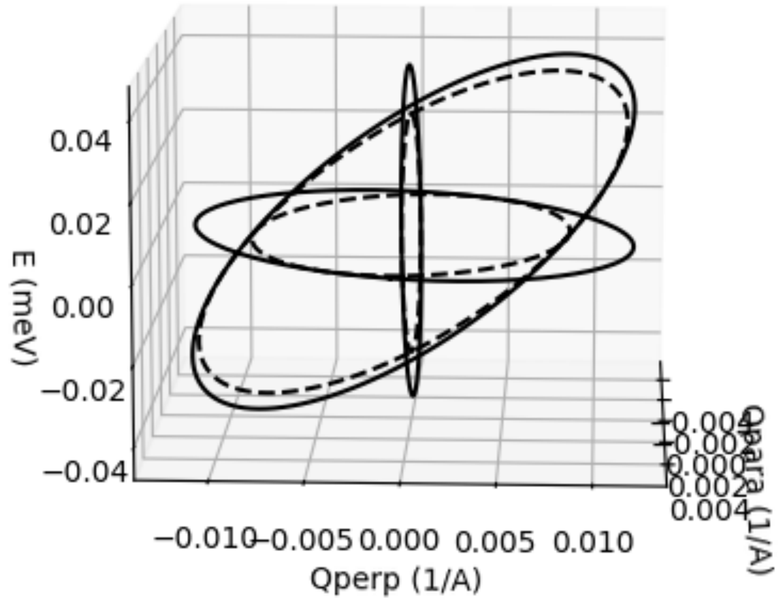
- Fit LG and TR scans. Exclude ones having mismatched background/ height
- Plot FWHM vs. Q. Exclude bad peaks.
- Tune instrument/sample parameters
- Remove the Lorentz factor
- Ready for refinement





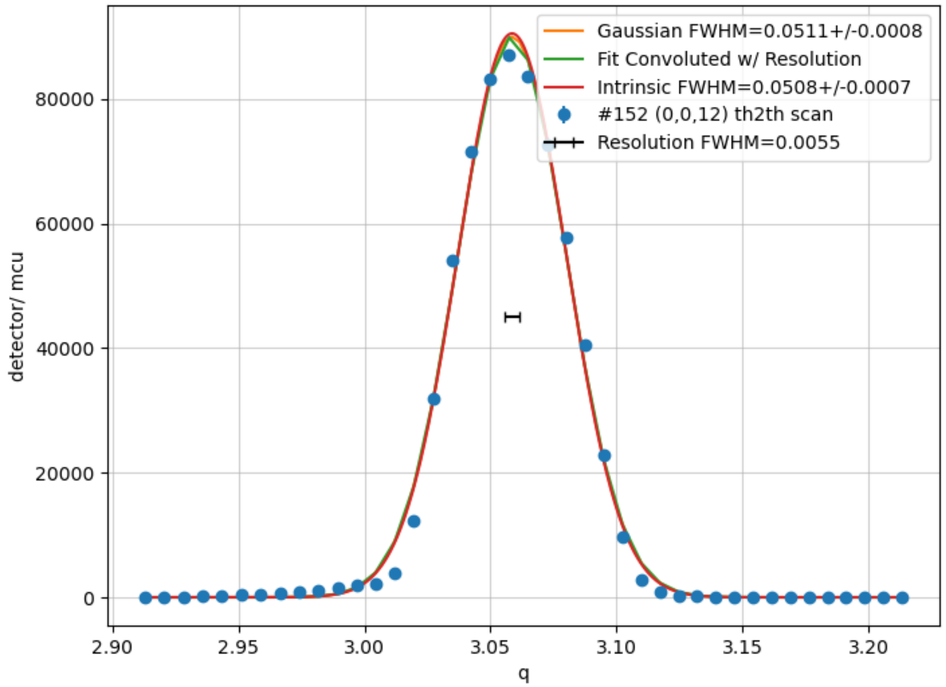
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Deconvolution of Resolution To Get Correlation Length

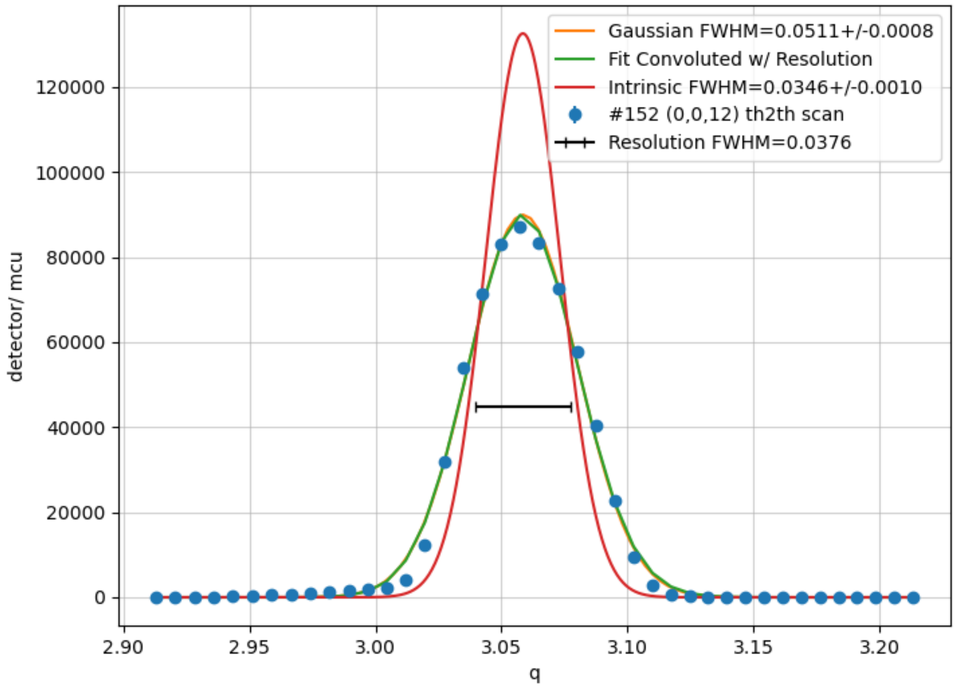
- $$I(Q_0) = \int_{-\infty}^{+\infty} \mathrm{d}Q\, S(Q)R(Q_0 - Q) = \int_{-\infty}^{+\infty} \mathrm{d}X\, S(Q_0 - X)R(X) \sim \int_{-3\sigma}^{3\sigma} \mathrm{d}X\, S(Q_0 - X)R(X)$$



- Sharp resolution
- Measurement is the same as the intrinsic signal

Deconvolution of Resolution To Get Correlation Length

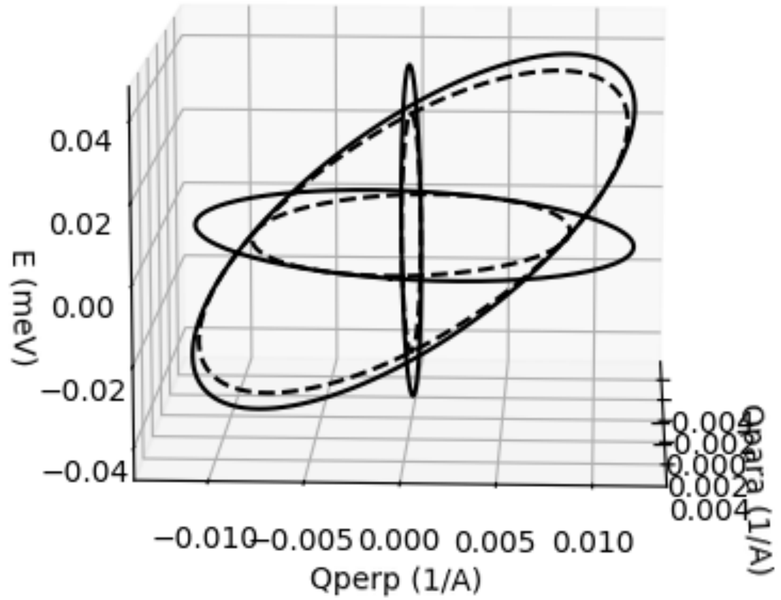
- $$I(Q_0) = \int_{-\infty}^{+\infty} \mathrm{d}Q\, S(Q)R(Q_0 - Q) = \int_{-\infty}^{+\infty} \mathrm{d}X\, S(Q_0 - X)R(X) \sim \int_{-3\sigma}^{3\sigma} \mathrm{d}X\, S(Q_0 - X)R(X)$$



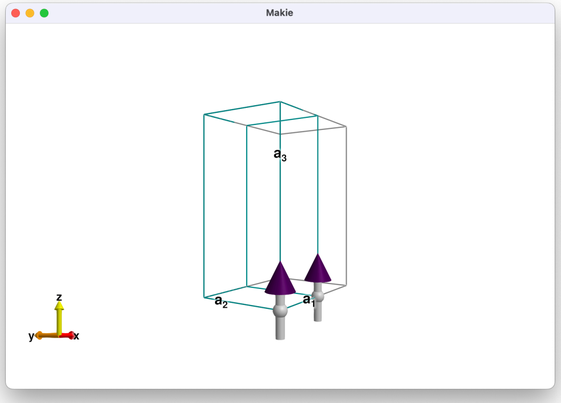
- Broad resolution
- Blue points/Orange curve: Exp FWHM ~ 0.0511Å^{-1}
- Black bar: Resolution FWHM ~ 0.0376 Å^{-1}
- Red curve: Intrinsic FWHM = $\sqrt{0.0511^2 - 0.0376^2} = 0.0346$ Å^{-1}
- Green curve: Fit result of the intrinsic signal convolved with resolution

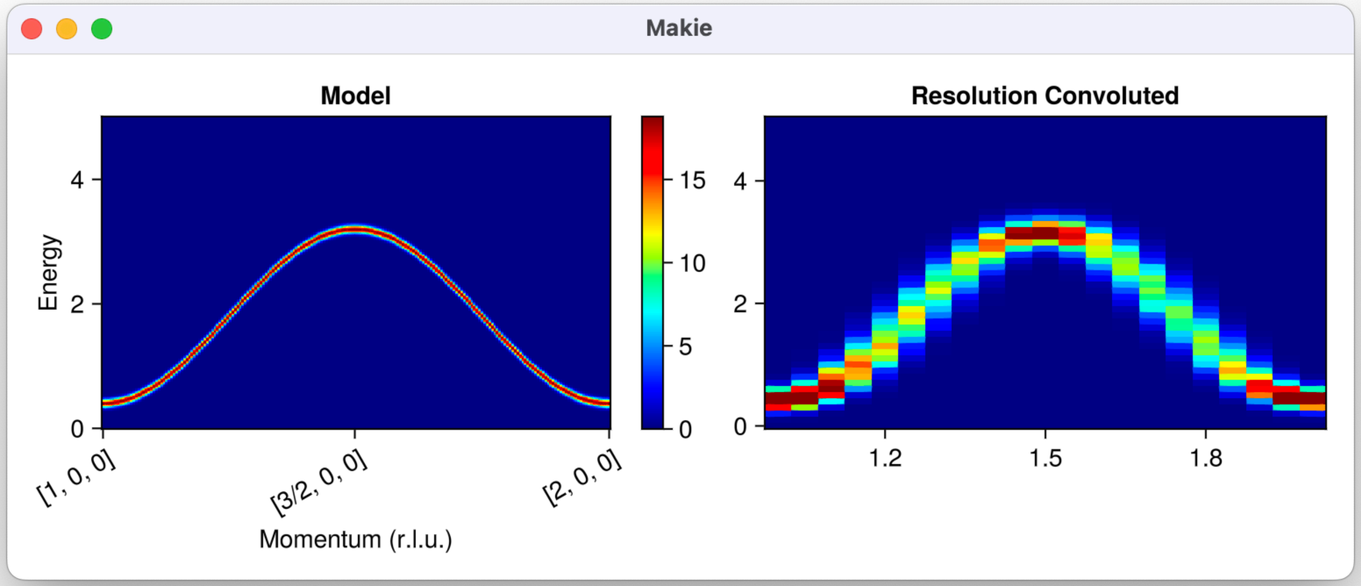
Contents

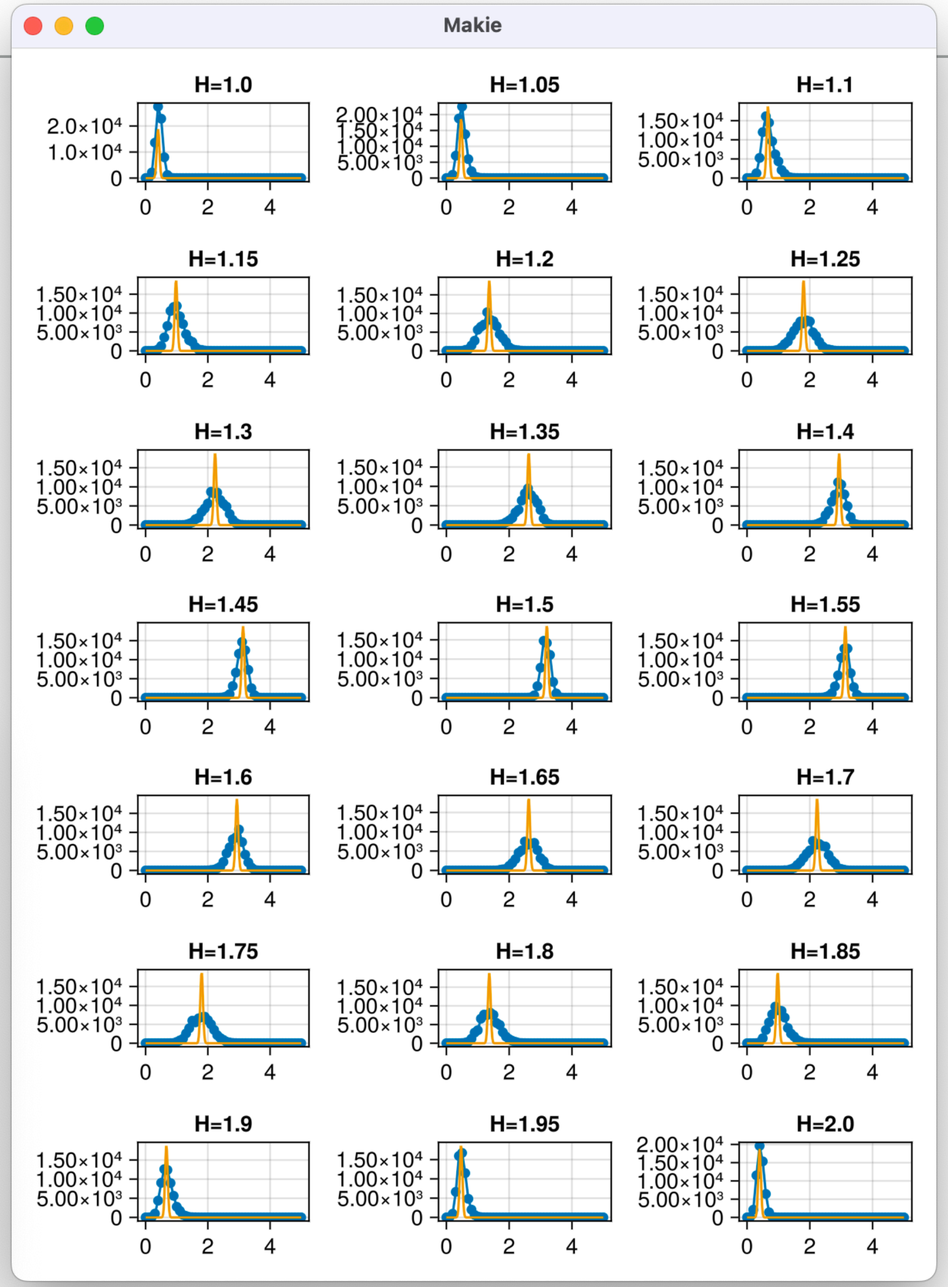
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1D FM Chain


Makie
a3
a2
a1
z
y
x


Makie
Model
Resolution Convoluted
Energy
Momentum (r.l.u.)
[1, 0, 0]
[3/2, 0, 0]
[2, 0, 0]
1.2
1.5
1.8


Makie
H=1.0
H=1.05
H=1.1
H=1.15
H=1.2
H=1.25
H=1.3
H=1.35
H=1.4
H=1.45
H=1.5
H=1.55
H=1.6
H=1.65
H=1.7
H=1.75
H=1.8
H=1.85
H=1.9
H=1.95
H=2.0

1D AFM Chain in Spin-Flop Phase

